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PA-141

APPRAISAL OF

LOWER KHALIS IRRIGATION PROJECT

IRAQ

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October 19, 1972

Europe, Middle East and North Africa Region

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CURRENCY EQUIVALENTS

US$1.00	=	Iraqi Dinars (ID) 0.329
ID 1.00	=	US$3.04
ID 1.00	=	1,000 fils

WEIGHTS AND MEASURES (METRIC SYSTEM)

1 hectare	=	4 donums, or 2.47 acres
1 kilometer (km)	=	0.62 miles
1 square kilometer (km^2)	=	100 hectares
1 meter (m)	=	3.28 feet
1 cubic meter (m^3)	=	1.31 cubic yards or 35.31 cu ft
1 million m^3 (Mm^3)	=	810 acre feet
1 cubic meter per second (m^3/sec)	=	35.31 cubic feet per second
1 liter per second (1/sec)	=	0.035 cubic feet per second

ABBREVIATIONS

O&M - Operation and Maintenance
C&F - Cost and Freight

FISCAL YEAR

April 1 - March 31

IRAQ

APPRAISAL OF
LOWER KHALIS IRRIGATION PROJECT

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This report was prepared by Messrs. C. M. Bolt, H. Bergendorff and T. H. Yoon (Bank), with contributions by Messrs. W. C. Hulsbos, W. B. Peters and E. Van Heurn (Consultants).

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APPRAISAL OF
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SUMMARY AND CONCLUSIONS

i. Agriculture plays an important role in the economy of Iraq, and irrigation is essential for raising crops throughout the greater part of the country. In its 1970-74 National Development Plan, the Government of Iraq has placed emphasis on the improvement and extension of irrigation systems.

ii. The Lower Khalis Irrigation Project would be a component of a major scheme to improve irrigation and drainage in the lower part of the Diyala basin in north eastern Iraq. The project would improve irrigation supplies to about 60,000 ha of land between the Diyala and Tigris rivers immediately north of Baghdad, and would include the construction of a 48-km extension of the Khalis main canal; distribution canals and pumping stations; a drainage system; flood protection works; roads; and levelling and buried field drainage of farm lands. It would also provide buildings, facilities, vehicles and equipment for the Administration in charge of the project area. Consultants would be engaged to design and supervise project works, to assist with studies of project cost recovery and marketing and storage facilities, as well as to advise on technical aspects of management.

iii. The project would include the construction of on-farm drainage and levelling before increased water supplies could be made available to farmers. These additional investments are designed to accelerate the adoption of improved irrigated farming practices, and to control soil salinization caused by inadequate drainage which is prevalent throughout much of Iraq. The project also provides for 180 km of rural roads and intensified agricultural extension and advisory services to give added encouragement to farmers to increase their crop output. Studies would be carried out to determine additional storage and marketing needs and the level and means of recovering the project costs.

iv. The estimated total cost of the project would be ID 25.6 million (US$77.8 million), which would be expended over the period ending in 1978. The estimated foreign exchange cost would be $40 million equivalent. The proposed loan of US$40 million would finance about 51% of the project cost and would cover all of the estimated foreign expenditure requirements.

v. The design of the project works and supervision of their construction would be assigned to a firm of consultants, who would also administer the bidding procedures and the certification of payments due to contractors.

vi. An autonomous Khalis Agricultural Administration (the Administration) would be responsible for the development of irrigated agriculture in the project area. A Director General would be the chairman of the Board of

Directors. The Ministry of Agrarian Reform would have the general responsibility for the Administration. To achieve a satisfactory rate of increase in farm production, consolidation and reallocation of project farm lands would have to be implemented by the Administration with the minimum of delay. Adequate measures to effect this would be taken by the Government.

vii. The planned extension of irrigation development on the Diyala river would eventually utilize all of the available water resources of the basin, depending on the extent of future abstractions in Iran. However, the Government would take all measures necessary to ensure that irrigation water supplies for the project would be adequate at all times.

viii. At full agricultural development of the project lands, the annual incremental net value of production would be about ID 5.5 million (US$16.5 million). The economic rate of return of the project would be about 12%. The project would lead to substantially higher farm incomes for about 10,000 farm families and nearly double the labor requirements in the project area. Other important project benefits include institution building, training and demonstration effects in technical and budgetary discipline in management of a public sector investment program which would contribute toward relieving some of the agricultural development constraints in Iraq.

ix. The project would be suitable for a Bank loan of US$40 million. The borrower would be the Government of Iraq and the loan would be repayable over 25 years with a grace period of 7 years.

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APPRAISAL OF LOWER KHALIS IRRIGATION PROJECT

I. INTRODUCTION

1.01 The Government of Iraq has requested a Bank loan of US$40 million to assist in financing the Lower Khalis Irrigation Project, located between the Diyala and Tigris rivers north of Baghdad. The project includes construction of canals and drainage systems, on-farm land preparation, flood protection, roads and buildings for development of about 56,700 ha for irrigated agriculture.

1.02 This would be the second loan in support of the development of Iraq's agricultural sector, after the first one for flood control which was made in 1950. A grain storage project has recently been appraised, and the Bank has agreed to assist the Government in undertaking a comprehensive study of the land and water resources of the Tigris-Euphrates basin in Iraq.

1.03 This report is based on the feasibility study prepared by the Government's consultants, Sir M. MacDonald and Partners and Hunting Technical Services (UK), with the guidance of the FAO/IBRD Cooperative Program. The project was appraised in November/December 1971 by Messrs. C. M. Bolt, H. Bergendorff, and T. H. Yoon (Bank), and Messrs. W. C. Hulsbos, W. B. Peters and E. Van Heurn (Consultants). This report was prepared by Messrs. Bolt, Bergendorff and Yoon, with contributions by Messrs. Hulsbos, Peters and Van Heurn.

II. BACKGROUND

2.01 General. Iraq covers an area of approximately 435,000 km^2 in the semi-arid zone of the Middle East, and has a population of about 9.7 million, which is growing at an annual rate of 3.3%. The per capita GNP in 1970 was estimated at US$330, and the income disparity between the urban and rural areas is wide. The rapid growth of the urban population, which is currently about 60% of the country's total, continues partly because of the lack of employment opportunity and the low income in the rural areas (about 25% of the urban income). The real annual growth of domestic output during the second half of the sixties was about 3.5%, substantially lower than the 10.6% annual growth rate during the first half of the sixties. The stagnation of the oil sector, modest agricultural output and the impact of nationalization and import restrictions on the industrial sector have been the primary causes of the poor recent growth performance. In 1970, however, real gross domestic output grew by 5.5% with an agricultural growth of 6.4%. Since early 1971, following the Tehran Agreement, oil sector revenues have increased from the 1969/70 level of $510 million to about $880 million in 1971/72. However, due to the nationalization on June 1, 1972 of the assets of the foreign Oil Company, the oil revenues in the short-run are likely to be less.

2.02 Agriculture in Iraq. Rainfall in Iraq is restricted to winter and varies from an annual average of 1,000 mm in the northern mountainous region to less than 100 mm in the south. Despite high rainfall, the mountains in the extreme north east are unsuitable for agriculture because of irregular topography. The piedmont zone to the east of the Tigris river is a rainfed agriculture area which supports a substantial livestock population. The main irrigated area of Iraq is in the plain between the Tigris and Euphrates rivers, south of the rainfed region. The northern and southern desert areas, west of the Euphrates river, support some livestock. Agriculture, after oil, is the second largest industry in Iraq and provides more than 50% of the employment. Even so, it accounts for only about one-fifth of the GDP. The sector is underdeveloped, and its productivity is among the lowest in the world. Foodstuffs comprised almost half of the consumer goods imported into Iraq during the past five years. Agricultural output in the past two decades has slowly increased and, during the period 1966 through 1970, the agricultural sector achieved an average real annual growth of 3.5%. This moderate growth partly contributed to the slowdown of the general economic expansion during the sixties. Agricultural production has also been affected by land reforms, poor investment planning and project management, and insufficient use of modern inputs, such as extension services and credit. The level of planned and realized agricultural investment has also been low, as agriculture accounted for only about 11% of total capital outlays, representing less than one-third of the planned sectoral investment for this period.

2.03 The Government has given high priority to agricultural development in the current 1970-74 National Development Plan, with the objective of achieving an annual agricultural growth rate of 7%. This was to be achieved primarily through an increase in the irrigated area and cropping intensity, and by increasing production of major cereals (wheat, barley and rice), cotton and oil seeds, and sugarcane. In view of the sectoral development constraints and in the light of past performance, the Plan targets appear optimistic. The major constraints to agricultural development in Iraq are: (a) shortage of irrigation water, soil salinity and drainage problems, and insufficient water distribution and control facilities; (b) lack of sufficient technical manpower and socio-economic factors impeding modernization of agriculture; and perhaps most importantly, (c) Iraq's limited capacity to plan and implement overall agricultural development strategy and associated investment projects, particularly in the important area of agricultural water resources development. The Government attaches a special significance to the proposed Lower Khalis Irrigation Project which would also provide valuable demonstration effects in overcoming some of these development constraints.

2.04 The Diyala Basin. At present, rudimentary irrigation systems in Iraq and Iran cover more than 280,000 ha in the basin, of which about 85% lie below the Jebel Hamrin gorge, where the Diyala river enters the alluvial plain of Central Iraq. The Lower Khalis project would be irrigated mainly from the Diyala river whose basin, shown on Map 1, covers a catchment of some 23,000 km^2, extending from the Iranian massif to the north of the

confluence with the Tigris near Baghdad. This area draws its water from the Diyala, and to a lesser extent from the Tigris. Once the project reaches full development around 1990, withdrawals from the Diyala would be about twice the present level, but pumping of the Tigris water would be reduced by about one-third as a result of the substitution of gravity supplies from the Diyala Weir.

2.05 The catchments of the Diyala and Tigris extend into Iran and Turkey respectively. Iraq does not have water treaties with either country, nor are there any plans which would lead to agreements being concluded in the near future. Both countries have been informed that the Bank intends to assist in financing the Lower Khalis Project. In reply, Turkey reserved the right to use the Tigris head-waters as it required and offered no comment on the project. Iran notified the Bank that it planned to divert the Diyala waters within its territories for irrigation and other purposes, thereby reducing the flow to Iraq. The effect of this possible reduction in the water supply to irrigation projects in the Diyala basin in Iraq, would depend on many factors like the cropping patterns adopted and the operation of proposed hydro-electric plants. Assurances were obtained during negotiations (para 4.15) that Iraq would take any measures necessary to ensure adequate irrigation water for the project area. Calculations show that even if the entire project water supply were met by an alternative source of pumping from the Tigris, the project would be economically viable with the rate of return at about 10% (para 7.04 and Annex 15).

2.06 The Lower Diyala region is potentially one of the most promising irrigated agricultural development areas in Iraq. The Government has already invested in major works, including the Derbendi Khan Dam, for flood control, and to a lesser extent, to improve irrigation and drainage in the basin. The dam was completed in 1962, and provision was made for a hydroelectric plant to be subsequently incorporated. Preparations are now being made to install generators with a capacity of more than 200 mw. The dam has enabled river flows to be increased in summer, but their use has been restricted by the inadequate capacities of the Lower Diyala canal systems. The Government has given priority to the enlargement of canals and construction of drains in the Upper Khalis area, on the right bank of the Diyala adjacent to the proposed project, and this work is due to be completed in 1972. Development plans include similar improvements on the left bank in the Lower Diyala region.

III. THE PROJECT AREA

3.01 General. The project area, shown in Map 2, lies between the Diyala and Tigris rivers, from Khalis town to the northern flood protection bund of Baghdad. It covers some 93,000 ha of land extending over four districts of the Baghdad and Diyala muhaphads (governorates), with a population of about 70,000 of which about 50,000 are directly agricultural. Transportation and communication between Baghdad and Khalis are good. The

distance of 60 km between the two points is spanned by two surfaced main roads, which run parallel on the eastern and western sides of the project area. There are few other roads in the area, but villages are served by cart tracks.

3.02 Climate. The summers are very hot and dry, with temperatures frequently reaching 45°C. Strong winds and low humidity cause high rates of evaporation and water use by crops. Rainfall, which occurs between November and April, averages only about 160 mm annually and is inadequate for crop growth even in the winter season, so that irrigation is necessary throughout the year. Night temperatures often fall to freezing point in December and January, which limits the choice of winter crops.

3.03 Soils and Land Classification. The topography of the project area is generally favorable for irrigation, except where the remnants of abandoned ancient canals and shallow basin depressions occur. The majority of the soils are fine-textured and some, particularly in the basin depressions, have relatively low permeability and poor internal drainage. The depth to groundwater presently varies from 1.5 m to 5 m in summer and rises about 1 m in winter. As the groundwater is very saline and evaporation high, there has been a buildup of salts in depressions and where the water table is close to the soil surface. Severe salinity occurs where internal drainage is poor and where crops have been regularly under-irrigated. Reclamation of most of these saline soils could be achieved by provision of adequate drainage. Soil fertility has been depleted by centuries of cropping, but yields would be improved by fertilizer applications after reduction of the salt content of the soil through leaching.

3.04 The land classification derived from surveys is adequate for forming a judgment on the feasibility of reclamation and irrigation development in most of the project lands. Prior to construction, a definitive drainage and land classification survey would be carried out under the project to enable on-farm works to be designed in detail. Within a gross area of 93,200 ha on the basis of present survey information, the net irrigable area which would be served by the project canals is expected to be about 61,700 ha (Annex 2). After taking into account the existing 5,000 ha of orchards, experimental and poultry farms which would continue to be served by the canal system but on which no further development would be carried out, farm lands with a total net irrigable area of 56,700 ha are expected to be fully drained and levelled under the project (Annex 2, Table 3).

3.05 Irrigation and Drainage. The project area at present receives water supplies through a rudimentary irrigation system supplied by the recently reconstructed Khalis main canal, and by pumps from the Diyala and Tigris rivers (Annexes 3 and 4). The Khalis main canal is fed from the Diyala weir, which is the principal water distribution point for the Lower Diyala region. In the Lower Khalis area, a net area of about 42,000 ha is now irrigated by the existing gravity canals and pumps. Due to the small capacity of the channels, the present irrigation intensities are very low, and siltation tends to cause further restrictions in water supplies. Field drainage works are practically nonexistent; however, the Khalis and Daudiya

outfall drains and pumping station, built as the first stage of a drainage system in 1962-64, have improved conditions in their immediate localities.

3.06 Farm Size and Land Tenure. Government records, which cover about one-third of the project area and are considered to be representative, indicate that 78% of the total holdings in the area consist of farms of less than 10 ha; about 20% range from 10-15 ha, and 2% of the holdings, covering less than 10% of the area, are larger than 15 ha. Expropriation of lands under the 1970 Agrarian Reform Law has been completed in the project area. Irrigated private holdings are now limited to between 37.5 ha and 100 ha depending on their location and fertility but, as indicated above, very few farms are larger than 15 ha. After expropriation, Government lands are distributed to smallholders and landless farmers, and families may receive up to 10 ha. A significant part of the expropriated lands in the project area has not yet been distributed, pending implementation of the project.

3.07 Agricultural Taxation and Rents. There are two laws which provide the basis for agricultural taxes in the project area. The Agrarian Reform Law of 1970 which applies to about 42% of the project area specifies shares (in cash or kind) of agricultural output to be distributed to each factor of production - land, water, cultivation and seeds, and administration. The percentage distribution of returns to these factors varies according to whether the land is rainfed or irrigated by gravity or pump. The Law of Agricultural Land Taxation which applies to non-Agrarian Reform lands provides varying land tax rates, ranging from 0.5% to 5% of agricultural production, depending on the method of irrigation and nature of ownership, but makes no provision for water charges. Accurate records on actual tax collection in the project area are not available but, as in many developing countries, some tax evasion is prevalent and Government support of public gravity irrigation water supplies seems to be a common expectation on the part of farmers.

3.08 Agricultural Production and Institutions. Farm production in the area is at present on a very low level, due mainly to the restricted crop intensities which average 40% in winter and under 10% in summer; and poor yields, resulting from the inadequate water supply, lack of drainage and associated soil salinity. The winter crops are barley in the salt-affected lands, and linseed, pulses and wheat on the better soils. In summer very limited areas of sesame, vegetables and cotton are grown. Orchards of citrus under sheltering date palms are raised mainly on the levee soils where pumped water supplies are available and drainage is good. The rearing of sheep, goats and, to a lesser extent, cattle is an integral part of farming.

3.09 At Baquba the Government has an extension service center which provides one officer for every 450 farmers, and operates a machinery pool mainly providing tractors for ploughing. The Government also operates a crop experimental station at Rashad, a poultry farm at Muradiyah and an Agricultural High School at Khalis. Cooperative Societies have been established in many of the villages in the area. These Societies supply short-term production input credit through the Agricultural Bank. Medium-term credit is also available for purchasing farm machinery and privately-owned tractors are already in common use in the area. Marketing is mainly in the hands of private traders for all farm produce, except vegetables which are sold directly by farmers in the Baghdad markets.

IV. THE PROJECT

A. Summary Description

4.01 The project would provide an efficient irrigation system to replace the existing canal layout of inadequate capacity between the Diyala and Tigris rivers south of Khalis and north of Baghdad. The net area irrigated would be increased from about 42,000 ha to 61,700 ha, the water delivery capacity would be correspondingly increased, and improvements would be carried out on 56,700 ha of farm lands (para 3.04). Main drainge systems would be constructed throughout the whole of the project area, in addition to a village road network and improved protection from Diyala floods. An Administration with jurisdiction over both the Upper and Lower Khalis areas would execute the project.

4.02 The project works would comprise:

(a) the construction of an irrigation distribution system, including the extension of the existing Khalis main canal by 48 km, two new pump stations, about 540 km of branch and distributary canals, and 2,900 km of water-courses;

(b) the provision of a land drainage disposal system including a pump station, about 440 km of open main drains, and 1,100 km of collector drains within the commanded area;

(c) the strengthening of the western flood protection embankment along the Diyala river;

(d) on-farm works including the installation of buried field drains and levelling of land in a net command area of about 56,700 ha;

(e) the construction of about 180 km of unpaved all-weather roads from villages to the main roads; and

(f) building works including housing, offices, workshops and other facilities for the Administration.

The project would also include the employment of consultant firms by the Administration; the purchase of vehicles and equipment for operation and maintenance of the project; strengthening of the agricultural extension service; and staff and farmer training. Studies would be made of marketing, storage and processing facilities required in the project area; and of means to meet the investment, operation and maintenance costs of the project. Details are given below and in Annex 5.

4.03 A critical factor which would affect the progress and cost of the project would be the close phasing of canal and drainage system construction

with on-farm development, which would require unrestricted access to farm lands in accordance with a schedule previously arranged between the Administration and farmers (para 4.10). The rate of increase in agricultural production would largely depend on the rapidity with which the resettlement of project lands can subsequently be completed by the Administration.

B. Detailed Features

4.04 Irrigation and Distribution System. The new reach of the Khalis main canal, 48 km in length, and with a head capacity of 53 m^3/sec, would extend from the end of the canal serving the Upper Khalis area to the north of the project and would be supplied from the Diyala river. The Khalis main canal would serve about 50,000 ha including project Development Areas I and II as shown in Map 2. Development Area III of about 7,000 ha in the southwest of the project area would be supplied from the Tigris river by new pump stations at Rashidiyah north and south. Distributary canals would supply several watercourses serving blocks of land (watercourse units) varying in size from 40 to 75 ha, depending on the type of farm system. There would be about 850 watercourse units within the project, and the watercourses and farm outlets, to be constructed by contractors, would supply farm field channels constructed by the farmers. The structural and hydraulic design of the irrigation system of unlined canals would be based on internationally accepted standards, and would have sufficient capacity to provide for variations in cropping patterns which can be expected to occur.

4.05 On-farm Development. Land levelling is an important requirement for the improvement of water distribution, irrigation efficiency and control of soil salinity, leading to higher agricultural output. To encourage the attainment of reasonable production levels as soon as possible, land levelling would be performed by contractors throughout each watercourse unit. Buried field drains would be installed in most farm areas to maintain satisfactorily low groundwater levels and soil salt contents, following more intensive irrigation. The field drains would be installed as part of the project works. The drains would each be 300 m long, and spaced at 100 to 200 m, depending on the soil permeability and the cropping system. Continuous, slotted plastic pipes of 100 mm diameter and a surrounding gravel filter would be placed in trenches excavated by machine, and would be designed to discharge freely into open collector drains. Preconstruction surveys would be carried out according to guidelines acceptable to the Bank to determine detailed drainage and land levelling requirements and costs, and to delineate the area to be developed under the project in detail.

4.06 Drainage Disposal System. Open drains would discharge flows from the buried field drains to the rivers through four drainage networks, each

consisting of open collectors, main drains and an outfall (assisted by pumps at Safia and Daudiya) with depths ranging from 2.5 to 3.5 m and having a total length of about 1,450 km. The collector drains would be about 1.5 km long and spaced 600 m apart. As they would form the limits of the water-course units, these deep open channels would establish revised farm and village boundaries, and assist in land redistribution procedures.

4.07 Flood Protection. The operation of Derbendi Khan dam for control of floods in the Diyala river will limit discharges below Diyala weir to about 3,000 m^3/sec except in abnormal circumstances which could occur with a frequency of once in 200 years or more. To ensure that the project farms in the vicinity of Rashad are protected from more frequent floods of between 2,000 and 3,000 m^3/sec, the project works would include the extension and raising of the flood embankment on the western bank of the river (Annex 5).

4.08 Buildings and Infrastructure. Administrative offices, staff housing, workshops and other facilities would be built at the Administration's headquarters in Baquba and at project Divisional Centers. Some 180 km of unpaved all-weather roads would be constructed to link all villages in the project area with the main roads. Each farmer would have access to his farm from the nearest village by earth roads adjacent to drains or canals. Warehouses for farm inputs such as seed and fertilizers and some facilities for marketing and storage of farm products already exist in the project area. Additional facilities would be required as the project develops, and the Administration would prepare a study of the need for further storage, marketing and processing facilities. A citrus nursery would be established at Baquba as soon as practicable to provide planting material. A trial and demonstration center is also urgently required for extension staff training. The Administration and its consultants would prepare plans for these by the end of 1972.

4.09 Consultant Services. Engineering and agricultural consultants would be employed to carry out a preconstruction survey, prepare detailed designs for the major civil and on-farm works, administer and supervise contracts as well as to advise and assist the Administration in the management and operation of the project. Government is considering proposals from consulting firms based upon terms of reference which were discussed and agreed between Iraq and the Bank. Agreement was reached during negotiations that the final terms and conditions would be mutually satisfactory to the Bank and the Government.

4.10 Access and Land Reallocation. The extensive reconstruction of the irrigation system and the on-farm works could cause considerable local dislocation and interruption of farming. To enable the work to be phased and carried out expeditiously, it would be necessary for contractors to have unhampered access to lands. Some landholdings and inhabitants would have to be relocated in order to meet the requirements of the project's irrigation and drainage layouts and farming systems. The Government already has powers to effect such changes in Agrarian Reform lands. During negotiations it

was agreed that timely measures to consolidate and reallocate private landholdings would be taken by the Government as necessary.

C. Construction Schedule

4.11 The project would be completed by March 1978, as shown in the schedule given in Annex 6. Buildings and facilities for the Administration would be constructed under a contract scheduled to start in the fourth quarter of 1973. The works required in each of the three Development Areas would be carried out under separate comprehensive contracts, commencing with Area III towards the second quarter of 1974. In Areas I and II, the main and branch canals would be constructed with temporary works incorporated to maintain existing irrigation supplies so as to minimize disturbance to farmers. The collector drains, watercourses and on-farm works would be carried out under each contract according to a schedule of seasonal "freeze" periods in which groups of watercourse units would be kept free of crops so as to expedite construction. Irrigation under the new system would start on about 2,500 ha of Area III in mid-1975, and would extend to the whole project area by 1978.

D. Water Demand and Supply

4.12 At full development, from about 1990 onwards, about 1,060 Mm^3 of irrigation water would be required annually for the projected cropping patterns and intensities on the 56,700 ha to be developed under the project and for the 5,000 ha of existing orchards and other nonproject areas (para 3.04) which would also be supplied by the canal systems. Further details are given in Annex 3.

4.13 Under the project, as much as possible of the area (85%) would be served by canals fed with gravity supplies from the Diyala river through the Khalis main supply canal, in preference to the more costly alternative of pumping from the adjacent rivers. Only 15% of the area which could not be commanded by the gravity system would be supplied from the Tigris by pumping, and would require about 190 Mm^3 of water annually at full development. The annual demand from the part of the project area to receive water supplies by gravity would be about 870 Mm^3, and to meet project peaks in summer the capacity of the Upper Khalis reach of the Khalis main canal would eventually have to be increased by 12 m^3/sec. During negotiations, assurances were obtained that the Government would take measures as necessary and at its own cost, to increase the capacity of the Khalis main canal upstream of the project area so as to meet the future peak irrigation water demand of the project.

4.14 Under existing conditions, Diyala river flows could be regulated by Derbendi Khan dam so that in 75% of years there would be sufficient discharge at the Diyala weir (3,620 Mm^3/yr) to meet at least 90% of the

projected annual irrigation requirements of the whole of the Lower Diyala region at full development. This would be acceptable, as the frequency and extent of probable shortages would be of the order generally allowed for in irrigation projects.

4.15 If the additional hydroelectric power plant which is to be installed at Derbendi Khan were to be operated to meet peak energy demands, re-regulating storage would be required downstream of the dam, e.g. at Jebel Hamrin. However, if the power plant is operated continuously to meet substantial base load requirements or should all the irrigation developments presently being planned by Iran upstream of the dam be implemented, serious water shortages for irrigation needs in the project region could occur unless Iraq resorted to alternative sources of water supply. The Government recognizes this and, during negotiations, agreed to take, at its own costs, all measures needed to ensure an adequate water supply at all times to meet the irrigation needs of the project. Government plans to carry out a comprehensive study of land and water resources of the Euphrates-Tigris basin in Iraq, under which the 1960 Diyala basin plan would be reviewed from the viewpoint of determining the areas in the basin that could be developed and the alternative means of ensuring irrigation water to the Lower Diyala region including provision of additional storage by the construction of Hamrin dam, and pumping from the Tigris (see Annex 4). For this study some technical assistance from the Bank has been requested. Since the Diyala weir serves areas other than those under the jurisdiction of the project Administration, assurances have been obtained that the weir would be operated and maintained to ensure adequate supply of water at all times for the project area. In evaluating the project, it was assumed that the alternative source of water, if required, would be the Tigris, since this is a feasible and straightforward solution, and that the additional pumps would be capable of entirely replacing the gravity supply from the Diyala river to the Lower Khalis area.

E. Cost Estimates

4.16 The total expenditure required to complete the project by March 1978 is estimated to be ID 25.6 million (US$77.8 million), with a foreign exchange element of US$40 million equivalent (51%), as detailed in Annex 7 and summarized below:

Item	Local Costs	For-eign	Total Costs	Local Costs	For-eign	Total Costs	Foreign Expenditure
	---ID Million----			----US$ Million---			%
Civil Works:							
Irrigation system	1.4	1.3	2.7	4.3	4.1	8.4	49
Drainage disposal system	2.5	3.0	5.6	7.7	9.2	16.9	54
Flood protection works	0.2	0.2	0.4	0.6	0.7	1.3	54
On-farm development	1.7	2.6	4.3	5.2	7.8	13.0	60
Roads	0.9	0.4	1.3	2.7	1.1	3.8	29
Buildings & general items	1.4	0.4	1.8	4.1	1.1	5.2	21
Sub-total	8.1	7.9	16.1	24.6	24.0	48.6	49
Administration & Engineering:							
Project Administration	0.5	-	0.5	1.5	-	1.5	0
Engineering & agricultural consultants	0.8	1.5	2.3	2.6	4.7	7.3	65
Maintenance equipment	0.1	0.7	0.8	0.4	2.0	2.4	84
Land acquisition	0.1	-	0.1	0.3	-	0.3	0
Sub-total	1.5	2.2	3.7	4.8	6.7	11.5	58
Contingencies:							
Physical (15%)	1.2	1.2	2.4	3.6	3.7	7.3	50
Price escalation (5% pa)	1.5	1.9	3.4	4.8	5.6	10.4	54
Sub-total	2.7	3.1	5.8	8.4	9.3	17.7	53
Total Project Costs	12.4	13.2	25.6	37.8	40.0	77.8	51

The cost estimates are based on late-1971 bid prices for irrigation and drainage works, roads and buildings. On-farm drainage works were estimated from costs of similar work being carried out in Iraq, Europe and North America, modified to allow for the use of new techniques and materials. Equipment costs were based on recent prices for imported vehicles and plant. The costs include buildings and staff salaries for the administration of Lower Khalis irrigation areas during the project period. Contingency items have been included for possible increases in construction quantities of 10-20% (averaging 15%), depending upon the preparation status of the estimates, and for increased costs due to price inflation assumed to occur at the rate of 5% per annum on both local and foreign cost items. The cost estimates are shown net of taxes.

F. Financial Plan

4.17 The project would be financed as follows:

	ID millions	US$ Millions equivalent	% of total
IBRD loan	13.2	40.0	51
Government of Iraq	12.4	37.8	49
Total	25.6	77.8	100

The proposed Bank loan of US$40 million would finance all of the foreign expenditure cost of the project. The Bank's contribution would be about 51% of project cost. The Government would finance the balance.

4.18 The estimated annual cost schedule is given in Annex 7 and summarized below.

Financial Year (Apr-Mar)	1972/3	1973/4	1974/5	1975/6	1976/7	1977/8	Total
I. D., millions	0.2	1.2	7.3	8.7	5.9	2.3	25.6
US$ millions equivalent	0.6	3.6	22.1	26.5	18.0	7.0	77.8

The Administration's annual construction schedule and the associated expenditure estimates would be reviewed by the Bank each year.

G. Procurement, Disbursement and Audit

4.19 Procurement. Vehicles and equipment needed for the supervision, operation and maintenance of the project by the Administration (Annex 7), would be procured after international competitive bidding on a C&F basis in accordance with the Bank's Guidelines except that, following the Government's established practice, a notice of intent to call for bids would be internationally advertised, and each invitation to bid would be published only in local papers. The civil works would be divided into four construction contracts, one for buildings and facilities for the Administration and three covering the whole of the project works within each of the Development Areas. The civil works contracts would be awarded after international competitive bidding fully in accordance with the Bank's Guidelines. No construction other than routine maintenance and repair work would be carried out by the Administration's forces.

4.20 Provision would be made for consideration of bids from Iraqi firms assembling or, at a later stage, manufacturing maintenance equipment, workshop machinery, vehicles or spare parts, and a preference of 15% or the applicable tariff whichever is the lower, would be given to local manufacturers. There are no preferential tariff agreements between Iraq and other member countries which would affect the evaluation of bids.

4.21 Disbursement. The Bank would finance 100% of foreign expenditures made under civil works contracts awarded to foreign contractors, or 49% of total expenditures (representing the foreign expenditure component) under contracts awarded to local contractors. Foreign expenditures for mobilization and plant advances made to civil works contractors, as certified by the engineering consultants and approved by the Administration, would also be financed by the Bank. The Bank would also finance either the C&F cost of imported maintenance equipment, workshop machinery, vehicles and spare parts for the project, or the foreign expenditures in the case of contracts awarded to local firms assembling such items in Iraq, or the ex-factory cost if they are manufactured in Iraq to a substantial extent. Payments of foreign expenditures for consultants services would also be eligible for financing from the proceeds of the Loan.

4.22 Auditing of Accounts. Each department of the Government maintains budgets and accounts which are subject to internal audit. Assurances were obtained during negotiations that certified satisfactorily audited annual financial statements of the Administration, including auditors' reports, would be forwarded to the Bank within six months of the end of each fiscal year.

V. Organization and Management

A. Project Organization

5.01 Agricultural development in Iraq is under the direction of the Planning Board and the Higher Agricultural Council. The Planning Board is responsible for establishing priorities, allocating funds and authorizing projects of all types, while the Council implements agricultural projects and exercises certain executive powers of the Planning Board, such as the authorization of awards of large contracts. The Council directs and coordinates the functions of the ministries involved in agricultural projects. In most ongoing projects the Ministries of Irrigation, Agrarian Reform and Agriculture are each responsible for various phases of project implementation. The Government recognizes that for some schemes it is vital to ensure a high degree of direction, integration and control of the wide variety of activities and disciplines involved. The Government has therefore passed the Agricultural Administrations Law (Annex 9), which provides for the establishment, by Presidential decree, of administrations for individual projects.

5.02 Following the passage of this law, a Presidential decree has established the Khalis Agricultural Administration (the Administration), with adequate powers and jurisdiction over the Upper and Lower Khalis areas. The Administration is responsible to the Ministry of Agrarian Reform and is a semi-autonomous agency of the Ministry, to be administered by an executive board under the chairmanship of a full-time and qualified Director General. The Director General of the Administration has been appointed.

5.03 Although the enabling legislation allows for the construction and other works to be executed by other government agencies such as the Ministry of Irrigation, the Government has confirmed that the ultimate responsibility for all such works would remain that of the Administration. As noted above, the jurisdiction of the Administration extends over Upper and Lower Khalis. This is necessary for unified control of maintenance, operation and water distribution from the Khalis main canal. The headquarters facilities and vehicles to be provided under the project would be adequate for administration of the development of both the Upper and Lower Khalis areas.

5.04 The Administration would establish an office for project development planning and an administrative office at the headquarters at Baquba, and a divisional services office at Bani Saad, the divisional center for the project area, as follows:

(a) The Development Office would consist of three sections:

(i) The functions of the first section would be the planning, survey, coordination, monitoring and evaluation of the projects within the Administration's jurisdiction.

(ii) The second section would be responsible for the preparation and execution of all construction contracts, including the detailed design and supervision of the project works. This work would be assigned, as at present, to the engineering consultants, who would provide the necessary staff.

(iii) The third section would administer the redistribution of lands and resettlement of farmers within the project. The section's work would include surveys to up-date the land tenure registry, drawing up new village and farm boundaries on the basis of the designed irrigation and drainage layout, and the registration of lands of new owners.

(b) The Administrative Office would be responsible for recruitment and staff relations, stores including procurement of equipment, and accounts including applications for disbursement from the loan.

(c) The Divisional Services Office at Beni Saad would be responsible for agricultural extension and cooperative services, arranging for farm credit, the collection of revenue, water distribution, and the operation and routine maintenance of project works after their construction.

5.05 Agricultural Extension. The Agricultural Section of the Divisional Services Office would provide extension officers to advise farmers on irrigation techniques, salinity control, new cropping patterns and more intensive farming than has previously been practiced in the area. One officer would be assigned to about 100 to 200 farmers. To support the extension service, a demonstration farm would be established, at which in-service training for staff would be provided by the Administration.

5.06 Cooperatives. Farmers' service cooperatives would be formed at an early date in each of the 32 major villages in the project area, and would be the focal points for the supply of agricultural inputs, machinery hire, and the collection of farm produce for marketing. The Cooperatives Section of the Divisional Services Office would provide supervisory officers for each village unit who would coordinate the requirements for bulk purchasing, distribution and marketing, and liaise with the Agricultural Section with regard to farmers' input and storage needs. A study of future requirements would be carried out under the project.

5.07 Credit. Short-and medium-term agricultural credit requirements would reach about ID 2.5 M (US$7.5M) in 1980. Sufficient credit to meet the needs of project farmers would be available through the Agricultural Bank, which plans to establish a branch in the project area and maintain close liaison with the Administration. Applications for credits would be made with the assistance of the extension service and the cooperative officers.

B. Operation and Maintenance

5.08 The Engineering Section of the Divisional Services Office would be responsible for the maintenance of the entire drainage system including buried field drains, for the operation and maintenance of project pump stations, irrigation works and all maintenance equipment, as well as for water distribution to distributary canal heads. Below this point, distribution would be arranged by canal-users' associations formed by farmers on each distributary unless equipment is required. The Engineering Section would be responsible for maintenance of the village road system in the project area.

5.09 The satisfactory regulation of water supplies from the Diyala river through the Khalis main canal would be of direct importance to the project during the construction and development stages. The Administration will also be responsible for the operation and maintenance of the whole of the Khalis main canal, the entire irrigation and drainage system in Upper and Lower Khalis, and for the future agricultural development of the areas under its jurisdiction.

5.10 Staffing. The Administration would be able to recruit its own staff in sufficient numbers, either directly or by secondment from Government ministries, or through consulting or other organizations. The Government has agreed to ensure that the Administration would be adequately staffed and efficiently managed at all times and recognizes the need for appointment

of key staff as soon as possible, preferably within six months of signing of the Loan Agreement. To assist with the activities of the Development Office, at least eight specialists would be engaged through the project consultants or directly by the Administration, for periods of 2 to 4 years (Annex 9). The engineering staff for supervision of the construction contracts would be provided by the project consultants if possible with Iraqis holding up to about 40% of the posts, if possible. About 70 agricultural officers would eventually be required for the project's extension service and at least four specialist agricultural extension advisors would be provided by the consultants to assist the Divisional Services Office for the first two years of the project's operation.

5.11 Programs. To enable the project to be planned and carried out with adequate funding, staffing and phasing of construction with on-farm development, it would be necessary to prepare detailed implementation programs for the various activities as soon as possible. Agreement was reached during negotiations that, with the assistance of the consultants when required, the Administration would aim to prepare the following programs for the project, within the following periods from the date of the signature of the loan agreement:

(a) an engineering development program within 4 months;

(b) a land settlement program within 6 months;

(c) an agricultural development and monitoring program within 9 months; and

(d) an administrative support program within 9 months.

It was also agreed that the Government would advise the Bank as soon as possible as to whether these dates are acceptable.

C. Recovery of Project Costs

5.12 In order to evaluate farmers' capacity to contribute toward the recovery of investment costs of the project, future farm incomes of representative farm types in the project area have been analyzed (Annex 12). This analysis suggests that significant revenues could be generated from the project without jeopardizing farmers' incentives. The illustrative project cost recovery schedule demonstrates how the entire investment cost, including on-farm development and operation and maintenance costs, within the project lifetime, at 3% interest, could be recovered. In this schedule, an effort was made to incorporate some degree of equitable distribution of

project benefits and efficiency of cost recovery by tapping a certain proportion of farm income in excess of the national average farm income. Consequently, as shown in Annex 12, the starting year of project charges toward the recovery of such costs varies according to each farmer's capacity to pay. The amounts of charges are progressive and redistributive in favor of poor farmers.

5.13 At this assumed level of recovery, the present value of government subsidy to farmers (measured by discounting the O & M costs, investments, including physical and price contingencies, and revenues over 50 years at 10%) amounts to about ID 14.5 million, and revenues would cover about 42% of costs. If the apportioned sunk costs without interest were included, the recovery rate would only be 34%. At this recovery rate, project charges would be approximately 40% of the average incremental net benefit. Considering the low farm income in the area "without project" which is projected at about half the national average farm income, such public subsidy may be justifiable. On the other hand, since it is very likely that the project would raise the average net farm income by almost five times at full development, the level of project charges indicated in Annex 12 would still leave adequate incentive to farmers.

5.14 While the Government has no detailed plans for cost recovery, the future need to apply some measure which would help equitable distribution of farmers' income as well as public subsidies resulting from this project has been recognized and the Government has stated its intention to generate a reasonable level of revenue from the project to cover at least the direct operation and maintenance costs, starting as soon as practicable. Under existing laws, the Government is empowered to collect a substantial fraction of farm production, depending on how the land is irrigated and whether the land is agrarian reform land or privately owned. While a comprehensive policy review on this subject has already been undertaken by the Government on a national scale, the Government has agreed to include, as a part of the project, the study of cost recovery for this specific project.

VI. PRODUCTION, MARKETING AND FARMERS' INCOME

A. Agricultural Production

6.01 The largest part of the project area, covering some 45,700 ha, would consist of 8-ha and 4-ha farms with traditional mixed arable field crops, forage and vegetables. The 8-ha farms would be located in areas with relatively low population density and the 4-ha farms in areas of higher population density. Farms specializing in vegetables and citrus would be located in the more densely populated part of Development Area II because of its soil characteristics and proximity to Baghdad. Composition of each project farm type and its location as shown in Map 2 is presented in the following table:

Farm Types and Crop Area

Farm type	Development unit	No. of farms	Area Under Each Crop					Total area
			Field crops	Vegetable specialist	Vegetables on mixed farms	Citrus	Forage	
			ha					
8-ha mixed arable	I(a,b,c) II(a)	4,670	5.0	-	0.5	-	2.5	37,370
4-ha mixed arable	III(a,b)	2,090	2.5	-	0.5	-	1.0	8,350
3-ha citrus	II(b)	720	-	0.5	-	2.0	0.5	2,160
3.6-ha vegetable	II(c)	2,440	-	2.0	-	-	1.6	8,870
Total		9,920	28,580	5,240	3,380	1,440	18,110	56,750

6.02 The present cropping intensity on the arable land in the project area is about 40% in winter and 10% in summer. At full project development in the 15th year, cropping intensities of arable crops are expected to increase to 90% in winter and 40% in summer. Annual cropping intensities are expected to rise from 90% in the first year of irrigation to 145% for vegetables on vegetable specialist farms, but for vegetables on mixed arable farms intensities would be about 20% lower. Citrus would be fully planted on 1,440 ha in 3 years. Details of crop intensities are presented in Annex 10.

6.03 The yields of arable crops are expected to increase during a 15-year development period by an average of 200-300% of the current yield level, whereas the rate of yield increase for that period, without the project, would probably be less than 1% per annum and may even deteriorate as a result of salinization. The project agricultural output at full maturity is estimated at ID 10.3 million in value compared to ID 1.6 million without the project. Yields, crop areas and total outputs are given in the following table.

Projections of Yields and Project Output

Crop	Present		Future		Incremental production at full development
	Yield (ton/ha)	Area (ha)	Yield (ton/ha)	Area (ha)	('000 ton)
Wheat	1.0	4,700	3.0	14,300	38.2
Barley	1.3	19,000	-	-	-
Linseed	0.7	920	1.5	5,700	8.5
Broad Beans	0.8	-	2.0	2,850	5.7
Winter Pulses	0.7	-	1.2	2,850	3.4
Sesame	0.4	-	1.0	2,850	2.9
Maize	1.0	-	3.0	1,400	4.2
Groundnuts	0.8	-	2.0	1,400	2.8
Seed Cotton	0.8	1,230	3.0	5,700	16.1
Citrus	-	-	3.0	1,440	36.0
Berseem-Barley Fodder	-	-	50.0	18,100	905.0
Maize-Sorghum Fodder	-	-	25.0	10,900	271.0
Vegetables	-	-	12.0	8,620	138.2
		25,850		76,110	

6.04 Timely and sufficient provision of production inputs and services including seeds, chemicals, farm machinery and supporting agricultural extension would be essential in order to achieve the target yields and intensities. There would be sufficient family labor available for the entire agricultural production, except for minor seasonal shortages on some individual farms which would be made up by surplus labor from others within the project area. Most of the seed for field crops would be domestically supplied but chemicals and some fertilizers may need to be imported. Citrus nurseries of the Administration would directly provide the planting materials for citrus farms. Agricultural credits for inputs, including farm machinery which is already widely used in the project area, would be obtainable from the Agricultural Bank. The detailed estimates of project agricultural inputs are presented in Annex 10.

B. Market Prospects, Commodity Prices, and Farmers' Income

6.05 Domestic market prospects are favorable for project output, particularly for high value crops such as vegetables and citrus due to the project's proximity to Baghdad and the import ban on citrus. The quantity of the project's agricultural output which could be internationally traded, including cereals, cotton and oilseeds, would contribute a negligible percentage to the total projected world trade volume of the respective items. Therefore, no noticeable effect on world trade market prices is foreseen.

6.06 Farmers' incomes in the project area are estimated on the basis of the projected farmgate price of the agricultural output. Prices of inputs are based on current market prices in the project area. Market prospects and the farmgate prices of project output used in economic and financial analyses, as well as their basis of estimation, are presented in Annex 11.

6.07 Annex 12 presents the detailed farm budgets for all farm types showing the annual flow of farm income and production costs as well as their repayment capacity toward recovery of project costs. Farm incomes at full development would be as follows:

	Citrus	Vegetable	Mixed 8-ha	Mixed 4-ha	Weighted Average with Project	Weighted Average without Project
	ID					
Gross value of farm output	1,300	1,120	1,170	640	1,060	180
Production costs excluding family labor	340	330	460	230	370	45
Net farm income before taxes and project charges	960	790	710	410	690	135
Project charges (See para 5.12 and Annex 12)	370	280	230	70	220	-
Disposable farm income after project charges	590	510	480	340	470	135

6.08 At full development the weighted average net farm income in the project area is expected to reach ID 690, compared to the projected income of ID 135 under "without project" conditions. After deducting

project charges used in the illustration (para 5.12 and Annex 12), the weighted average farm income in the area would be about ID 470. Even the poorest 4-ha mixed arable farm would be left with a net farm income of ID 340, higher than the projected national average farm income in Iraq.

VII. BENEFITS AND JUSTIFICATION

A. Economic Justification

7.01 The provisions of drainage, irrigation, on-farm development combined with other inputs and extension services under the project would bring systematic, intensive and modern agriculture into the net cultivated area of some 56,700 ha, which is currently subject to low cropping intensity and yields, with little prospect for improvement without the project. The major quantifiable benefit from the project would be the substantial increase in agricultural production which would raise farmers' income and increase employment opportunities. Among the intangible benefits which have not been included in the analysis are the demonstration effects of the project, institution-building and training, which would contribute to the acceleration of agricultural development in Iraq.

7.02 At full development the annual gross value of production for the project area is expected to reach ID 10.3 million, nearly six times the expected output without the project. After deducting the social costs of production and without costing the family labor inputs, the incremental value of the net annual social benefits attributable to the project would be ID 5.5 million.

7.03 The economic rate of return of the project is estimated at 12% (Annex 13). If the sunk costs proportional to this project without interest were included in the first project year's investment, the rate of return would be about 9%. Extensive sensitivity tests have been performed under various assumptions with respect to important variables in the benefit and cost models. The general conclusion of these analyses is that the economic rate of return of 12% is relatively insensitive to the changes in these variables within reasonable ranges, given the diversity of the benefit sources of this project. Detailed results of the sensitivity analyses are presented in Annex 14.

7.04 In view of the possible future water shortage in the Diyala basin, depending on the Iraqi and Iranian water resources developments, as discussed in para 4.15, the economic effect of supplementing the water supply by pumping from the Tigris River has been estimated. In this analysis various assumptions have been made regarding the development plans of the Diyala basin, allocation of apportioned pumping costs, and the level of upstream water resource diversion. Even if the entire project water supply were met by the alternative source of pumping from the Tigris,

the project would be economically viable with the rate of return at about 10%. The details of the economic analysis with respect to the pumping requirements are presented in Annex 15.

7.05 Unemployment and underemployment in the project area are severe. The entire incremental agricultural output from the project can be produced by utilizing this surplus labor. The details of the annual labor requirements and seasonal labor utilization with and without the project are presented in Annex 10, Fig. 1. Total annual project employment at full development is estimated at some 17,000 man-years, roughly doubling the employment volume projected without the project. The aggregate of employment during the construction period 1973-77 is estimated at some 18,000 man-years. The incremental output would undoubtedly stimulate further economic activities and employment for marketing, processing and supporting industries.

7.06 Because of the marked difference in farming systems with and without the project, an accurate quantitative account of distributive effects cannot be estimated. However, it is clear that practically all farmers would be substantially better off with the project. As shown in Annex 12, the average net farm income of every farm type would be significantly higher than average farm incomes without the project throughout the entire period. The project farm labor utilization would lead to relatively higher benefits to the smaller farmers with labor surplus. Furthermore, as illustrated in Annex 12, capital recovery schedules can incorporate progressive redistributive effects which would enable effective capital recovery and some degree of income equalization.

B. Ecology and Environment

7.07 There would be a general rise in the groundwater level throughout the project area as a result of increased irrigation. However, the installation of buried field drains and drainage disposal systems would virtually eliminate the possibility of waterlogging as well as considerably reduce soil salinity in the area to be developed. Winter temperatures are sufficiently low to inhibit the spread of schistosomiasis into the area. The extension and raising of the Diyala bund would reduce the risk of flooding near Rashad. Under reasonably efficient operation and agricultural practices, no significant negative effect on the ecology and environment would be expected.

VIII. AGREEMENTS REACHED AND RECOMMENDATIONS

8.01 During loan negotiations, agreement was reached on the following principal points:

(a) measures would be taken to provide the Government with powers to reallocate private landholdings in the project area as needed by the layout of the irrigation and drainage works (para 4.10);

(b) measures would be taken as necessary to increase the capacity of the Khalis main canal upstream of the project area to meet the future peak irrigation water demand in the project (para 4.13);

(c) The Diyala Weir would be so operated and maintained as to ensure adequate water supply to the project (para 4.14 and 4.15);

(d) to ensure that the supply of water to lands in the project area will be adequate at all times to meet the irrigation needs of the project, all such measures as may be needed would be taken, including the construction of any additional irrigation water facilities (para 4.15);

(e) a study would be made of means to meet the investment, operation and maintenance costs of the project; and necessary steps would be taken to generate revenue from the project as soon as practicable after its completion, to cover at least the operation and maintenance costs of the project (para 5.14).

8.02 The project would be suitable for a Bank loan to the Government of Iraq of US$40 million equivalent, repayable over 25 years, with a grace period of 7 years.

October 19, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

National Agricultural Output

Table 1

Output and Yield of Principal Crops in Iraq: 1964/65-1968/69

	Unit	1965	1966	1967	1968	1969
Dates:						
Output	'000 m tons	310.0	280.0	380.0	330.0	260.0
Yield	ton/ha	2.7	2.5	3.4	2.9	1.8
Wheat:						
Output	'000 m tons	1,006.0	826.0	860.0	1,371.0	1,189.0
Yield	ton/ha	0.6	0.5	0.5	0.7	0.6
Barley:						
Output	'000 m tons	860.0	832.0	855.0	931.0	1,250.0
Yield	ton/ha	0.7	0.7	0.8	0.8	1.0
Rice:						
Output	'000 m tons	198.0	182.0	308.0	325.0	284.0
Yield	ton/ha	1.7	1.6	2.2	2.3	2.0
Cotton:						
Output	'000 m tons	32.0	29.0	35.0	41.0	40.0
Yield	ton/ha	0.7	0.9	0.9	0.9	0.9

Table 2

Livestock Population and Slaughtering in Iraq: 1965-1969
(in Thousands)

	1965	1966	1967	1968	1969
Sheep:					
Population	11,040	11,850	9,720	10,800	12,000
Slaughtered	1,844	2,105	2,347	2,417	2,628
Goats:					
Population	1,845	1,800	810	900	1,000
Slaughtered	517	542	430	447	469
Cattle:					
Population	1,459	1,390	1,206	1,340	1,500
Slaughtered	276	263	279	289	300
Buffaloes:					
Population	225	380	405	450	500
Slaughtered	22	19	18	22	25
Camels:					
Population	202	217	205	225	250
Slaughtered	13	14	12	12	14

Source: IBRD Report No. EMA 35a
June 22, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Soils and Land Classification

Topography

1. The project area is generally very flat, with an average gradient of about 0.02%. The slope is mainly from the Diyala river to the Tigris river with an aspect from northeast to southwest. A principal topographic deficiency is the rough terrain that comprises remnants of villages and portions of ancient irrigation courses that have been abandoned, and costs for correction of this deficiency would be very high in many areas. Another deficiency is the widespread occurrence of depressions and microgullies consisting of relatively fine-textured and slowly permeable soils. "Gilgai", or channelized relief, is common to the basin depressions; however, this is not severe.

Soils

2. The soils are derived from alluvium originating in the upper reaches of the Diyala, and deposited in the area mainly through river flooding but also by irrigation deposition. The soils comprising the river levees and the areas of deposition adjacent to the water courses are coarse to medium textured and are layered. The lands at some distance from the water courses, particularly those in the basin depressions, consist of fine-textured so-called "Black Cotton" or "self-mulching" soils. These commonly have horizons in which the vertical and horizontal hydraulic conductivity values are low.

3. Moisture retention and the available water-holding capacity of the coarser textured soils are ample, and of the finer textured soils are ideal for irrigated agriculture. However, some of the fine-textured soils have layers in the root zone and to a depth of about 2 m whose bulk specific gravities are in the range 1.6 to 1.9; such values connote low permeability to water. The water intake and vertical and horizontal hydraulic conductivity rates are variable, ranging from moderate to very low values. In the basin depressions, some of these measured values are sufficiently low to indicate problems in maintaining a permanent irrigated agriculture. Many of the infiltration rates measured in the field leveled off to values of less than 0.1 cm per hour after 4 hours of testing and with accumulated values of only 2-3 cm of water during this period. Similar data was obtained in testing at the 20-40 cm depth.

4. Salt-affected soils are common to the area, and they contain mainly the sulfates and chlorides of calcium, magnesium, and sodium. However, most of them apparently are easily reclaimed through leaching and

drainage once irrigation water and drainage facilities are provided. The available laboratory data indicate that many of these are saline alkali soils as judged by their sodium adsorption ratio (SAR) and exchangeable-sodium percentage (ESP) values. However, they are calcareous and usually contain large amounts of residual gypsum. It is probable that the lime and gypsum precipitated in the soil profiles during salinization, even though the soils were high in exchangeable sodium and had high SAR values, for this can occur in slowly permeable soils. Upon dilution of the soil solution, gypsum dissolves, providing soluble calcium, and both the SAR and the ESP of the soils decrease. Since residual gypsum generally is present in excess of the exchangeable sodium, calcium would be supplied continuously and would displace the absorbed sodium during leaching provided drainage can occur. Thus, the high measured values of SAR and ESP are actually a chemical artifact, the result of salt accumulation and gypsum precipitation, and do not constitute alkali. Recent studies tend to dispel notions that soil structure stability is a problem in these soils. The factor critical to reclamation of the project lands is soil permeability and the effectiveness of the drainage to be installed.

5. The cation-exchange capacities range from about 15 to 25 milliequivalents per hundred grams. It is believed that these soils are low with respect to available nitrogen and phosphorous levels. This is to be anticipated in view of the climate and a very long cropping history without significant fertilizer inputs. The lands should respond to fertilizer applications, particularly those having good physical conditions, low salts, and adequate drainage.

Soil Survey and Land Classification Studies

6. The feasibility study is largely based on the land resources survey performed prior to 1958, which merely consisted of a reconnaissance soil survey; this was translated into a land classification. The principal criterion for differentiation between land classes was the salinity level at the time of survey. Some of the lands that were classed as nonarable because of salinity are now non-saline, and others are suitable for irrigation development, provided that water is available for leaching, and that drainage can be installed at reasonable cost. The land classification did not adequately cover topography deficiencies and land-leveling requirements. The 1958 survey was supplemented in 1970 with a drainage study, consisting mostly of soil permeability measurements made below drain depths (i.e. at 3 to 4 meters) where the water table was encountered. There are very few measurements for permeability made in the zone where the drains would be installed. In 1971 additional field and laboratory tests were made to characterize the soils with respect to their capacity to permit the movement of water and salts. The field permeability testing revealed a high incidence of slowly permeable sub-soils in some areas, whereas other areas had satisfactory rates of vertical and horizontal hydraulic conductivity. Although field tests showed that excess salts could be leached out of most soil profiles, many measurements of water intake rate approached critical values. Upon completion of the supplementary work, the results derived from the 1958 survey were revised and a new land classification map was prepared. The revised land classes were not well defined, they included

overlapping classes as there was little difference among some of the criteria used to distinguish between arable and nonarable land, and some of the lands were delineated on the basis of existing rather than on future conditions.

7. The existing land classification is, however, suitable for determining on a judgment basis the feasibility of irrigation development in most of the project area, but is not of sufficient scope and intensity of coverage to be used in preconstruction investigations, particularly in Development Area I. In the absence of a definitive and detailed land classification, an analysis of the available data has been made, and the results are shown in Table 1. Although the net area actually to be developed under the project would depend upon the results of a preconstruction survey which is required for construction, the most probable net area suitable for irrigation development would be approximately 61,700 ha, as shown in Table 2. Of this area, about 56,700 ha net would be selected for on-farm development under the project, distributed as shown in Table 3; the remaining 5,000 ha is already under orchard or other development, as indicated in the table.

Pre-construction Drainage and Land Classification Survey

8. The survey would establish the spacing of buried field drains and land levelling requirements, and hence the development costs, for each group of watercourse units throughout the project, according to the soil and groundwater conditions as encountered and to be anticipated when more intensive irrigation and drainage is applied under the project. As a result of the survey, lands which would require excessively costly land levelling and drainage would be selectively excluded. The survey would be carried out concurrently with the topographical and other necessary survey work prior to construction of the irrigation and drainage systems.

ANNEX 2
Table 1

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Soils and Land Classification

Gross Areas of Land/Soil Classes

Land/Soil Class	Development Area I	II	III	TOTAL
	(- - - - - Gross Area, ha - - - - - - -)			
Satisfactory categories:				
Class 1	1,300	6,300	5,070	12,670
Class 2	3,670	15,940	1,520	20,130
Class 3 (satisfactory)	2,280	7,540	1,700	11,520
Orchards	2,420	150	1,840	4,410
Satisfactory lands	9,670	29,930	10,130	49,730
Areas to be re-surveyed:				
Class 3 (re-survey)	13,090	13,000	1,420	27,510
Class 6 (re-survey)	5,950	1,500	400	7,850
Lands to be re-surveyed	19,040	14,500	1,820	35,360
Excluded areas				
Class 6 (excluded)	3,160	4,440	290	7,890
Settlements	-	210	-	210
	3,160	4,650	290	8,100
Total Gross Areas	31,870	49,080	12,240	93,190

October 17, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Soils and Land Classification

Probable Distribution of Arable Lands Within Project

Development Area	Gross Area Satisfactory Lands (ha)	Gross Area, Land to Be Resurveyed (ha)	Percentage Probability of Acceptance of Lands to be Resurveyed (%)			Gross Area, Range of Lands Which After Resurvey Would Probably be Acceptable (ha) (2) x (3)			Gross Area, Probable Range of Arable Lands in Project (ha) (1) x (4)			Net Area, Probable Range of Arable Lands in Project (ha) (5) x 0.88		
	(1)	(2)	(3)			(4)			(5)			(6)		
			Low	Medium	High	Low	Medium	High	Low	Medium	High	Low	Medium	High
I	9,680	19,040	35	50	65	6,660	9,520	11,420	16,340	19,200	21,100	14,350	16,900	18,500
II	29,930	14,500	50	65	80	7,250	9,430	11,600	37,180	39,360	41,530	32,750	34,600	36,550
III	10,120	1,820	70	80	90	1,280	1,460	1,640	11,400	11,580	11,760	10,000	10,200	10,350
Totals	49,730	35,360					20,410			70,140		57,100	61,700 [1]	65,400

[1] Most probable net culturable area within project canal command after deduction of 12% for infrastructure area; taken to Table 3, column (3).

June 22, 1972

ANNEX 2
Table 2

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Soils and Land Classification

Development Areas

Development Area	Farming System	Net Commanded Area (ha)	Net Area Not to Be Developed under Project (ha)	Net Area to Be Developed in Project (ha)
(1)	(2)	(3)	(4)	(5)
I(a)	Mixed arable	9,850	- 1/	9,850
I(b)	Mixed arable	3,110	1,160 1/	1,950
I(c)	Mixed arable	3,940	1,040 1/	2,900
	Subtotals	16,900	2,200	14,700
II(a)	Mixed arable	23,600	900 2/	22,700
II(b)	Citrus	2,150	-	2,150
II(c)	Specialist vegetable	8,850	-	8,850
	Subtotals	34,600	900	33,700
III(a)	Mixed arable	2,850	550 1/	2,300
III(b)	Mixed arable	6,250	1,350 1/	4,900
Tigris riverain	Mixed arable	1,100	-	-
	Subtotals	10,200	1,900	8,300
	Totals	61,700	5,000	56,700

1/ Existing orchards.

2/ Muradiya Poultry Farm.

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LOWER KHALIS IRRIGATION PROJECT

Water Demand and Supply

A. Water Demand

1. Consumptive use of water by project farms was estimated for each crop by months, on the basis of evaporation rates and coefficients relating to the stage of crop growth. The probable losses due to percolation of irrigation supplies below the root zone and to farm waste were taken into account. The proportion of rainfall which would be effective for plant growth was also allowed for, together with watercourse and canal seepage and operational losses, as outlined below.

Evaporation Data

2. Records of evaporation from a Class "A" pan at Baghdad Airport were obtained from 1966-69 inclusive and were as follows:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
mm											
71	94	170	223	356	436	501	453	301	202	118	62

3. The evaporation from a free water surface was computed using the Penman formula, from climate data recorded at Baghdad Airport, and indicated an average proportion of 0.7 of the Class "A" pan observations, as follows:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
mm											
50	84	146	198	260	309	332	298	213	133	69	43

Rainfall Data

4. Records of rainfall in Baghdad for the period 1937-1968 indicated the following monthly averages:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
mm											
25.5	28.0	26.5	20.1	7.4	0.1	-	-	0.1	3.5	20.3	24.9

5. When the monthly rainfall exceeded 20 mm, calculated crop water requirements were reduced by 65% of the monthly average rainfall. The effective rainfall was estimated to be as follows:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
mm											
17	18	17	13	-	-	-	-	-	-	13	16

Crop Consumptive Use

6. Using crop coefficients derived from data obtained in Iraq, Egypt and Syria, and given in Table 1, the consumptive use of individual crops was calculated by the Penman method, and checked by the Blaney-Criddle method, which gave results 8-10% lower than Penman. The values of the net monthly irrigation requirements, after deduction of the effective rainfall, are given in Table 2.

Cropping Pattern Irrigation Requirements

7. To assess the farm irrigation requirements, the total irrigation requirements for the combination of crops in each cropping pattern were calculated, and are shown in Table 3, before allowance for field losses.

Losses

8. The following allowances were made for operational losses in estimating the total irrigation water requirements for the project:

Field losses (deep percolation and waste), 25%
(Equivalent to a field efficiency of 75%)

Watercourse losses (seepage and evaporation), 10%

Canal losses, 2.5 m^3/sec per million m^2 of canal wetted perimeter, (13-25%).

Project Water Demand

9. Based on the cropping pattern in each farm system, the net area in each Development Unit and the field and transmission losses, the monthly water requirements for the project at full development were estimated and are shown in Table 4. Separate amounts have been derived for areas to be supplied by gravity from the Diyala river and by pumping from the Tigris. The maximum monthly rate of demand from the Khalis main canal would average about 47 m^3/sec, with short-duration peaks up to 53 m^3/sec, and the maximum pumping demand from the Tigris would be about 11 m^3/sec as compared with the present rate of 19 m^3/sec.

B. Water Supply

The Diyala Basin

10. The Diyala river rises in the mountains of Iran and flows southwestwards until it joins the Tigris near Baghdad. Irrigation works dating back to 3,000 B. C. have been identified in the valley which, in more recent times, was the frontier between the Ottoman and Persian empires. The average flow of the Diyala is 178 m^3/sec, or only 6.6% of Iraq's water resources, but the river provided more than 10% of actual water use during 1960-69.

11. The Diyala basin, shown in Map 1, covers about 32,600 km^2, of which 17,500 km^2 lies in Iran. It comprises three distinct areas:

(a) The "upper catchment", above Derbendi Khan dam, of about 17,850 km^2. The dam, completed in 1962 at a cost of about ID 20 million (US$61 million), lies in a very narrow gorge in a steep, high ridge which isolates the upper mountainous part of the catchment from the foothills. The rivers, fed mainly by snow-melt, are perennial and, regulated by the dam, provide water for summer crops in the lower parts of the catchment.

(b) The "intermediate catchment," of about 12,800 km^2, between Derbendi Khan and the Jebel Hamrin, a ridge separating the foothills from the Mesopotamian alluvial plain. Five small perennial rivers - four rising in Iran - join the Diyala in this reach, but most of the runoff from the intermediate catchment is highly seasonal, characterized as "flash floods" immediately after rain.

(c) the "lower catchment," of about 1,900 km^2, between Jebel Hamrin and the Tigris. The rainfall in this area is less than 400 mm and runoff after rain is small. The Diyala weir, situated below a gorge in the Jebel Hamrin ridge, diverts water for irrigation of the alluvial plain of the Lower Diyala region.

12. There is remarkably little hydrologic information on the Diyala. The only long-term records are for the Diyala weir, which was built in 1919. Some figures quoted in feasibility and other reports are based on a very few season's records correlated mathematically with the 39-year series of observations at the weir. Other figures have been derived from rainfall recorded for 26 years at six rain gauges in the catchment.

13. Flow observations recorded at Diyala weir from 1931 through 1969 indicate that the average annual discharge above the weir is about 5,500 Mm^3. An analysis of the appropriate operation of Derbendi Khan dam for irrigation purposes shows that, if diversions in the intermediate catchment are controlled, the annual flow which can be expected to be equalled or

exceeded at the weir in 75% of years with the present extent of upstream irrigation development, would be about 3,600 Mm^3. The monthly distribution of flows which could be regulated at Derbendi Khan for irrigation downstream is are shown in the summary Table 5. The summary table also includes the monthly distribution of demand for Lower Khalis project as given in Table 4, as well as the total demand at Diyala weir from the Upper and Lower Khalis projects at full development in addition to the demand from the other Lower Diyala areas served at their present state of development. This level of demand could easily be met by the 75% probability flows expected at Diyala weir, as shown in the Table 5. Anticipated development of the Lower Diyala region would, however, give rise to increased water demand, mainly as a result of higher cropping intensities. At full development of the net area of 295,000 ha which could be commanded at Diyala weir, the 75%-probability river flows would be sufficient to meet 90% of the annual demand from the area if the whole of the shortage is distributed from April through September, resulting in about 15% shortage in the summer season. The related water demand for this condition is also shown in Table 5.

14. This analysis was based on the assumptions that releases for hydroelectric power production at Derbendi Khan dam would be given no priority over irrigation releases, and that the inflow to the reservoir would be unaffected by upstream developments. The effects of possible additional diversions for irrigation in Iran are described in Annex 4.

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IRAQ

LOWER KHALIS IRRIGATION PROJECT

Water Demand and Supply

Crop Cooefficients

Crop	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Wheat	1.00	0.90	0.75	0.65	0.55	-	-	-	-	-	0.55	0.80
Barley	0.95	0.85	0.70	0.50	0.35	-	-	-	-	0.45	0.75	1.00
Linseed	0.95	0.80	0.75	0.55	0.30	0.25	-	-	-	-	0.70	1.20
Broad Beans	1.05	0.60	0.45	-	-	-	-	-	-	0.35	0.60	1.10
Winter Pulses	1.20	0.90	0.60	0.45	-	-	-	-	-	0.30	0.50	0.95
Cotton	-	-	0.90	0.40	0.60	0.85	1.00	0.65	0.45	-	-	-
Maize	-	-	-	0.30	0.55	0.95	0.90	0.60	-	-	-	-
Green Gram	-	-	-	-	0.60	0.80	1.05	1.15	0.90	-	-	-
Groundnuts	-	-	-	0.25	0.40	0.65	0.85	0.90	0.75	0.50	-	-
Sorghum	-	-	-	0.55	0.90	1.15	0.80	0.45	-	-	-	-
Sesame (late)	-	-	-	-	0.20	0.50	0.85	0.80	0.45	0.35	-	-
Sesame (early)	-	-	-	0.20	0.50	0.85	0.80	0.45	0.35	-	-	-

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ANNEX 3
Table 1

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Water Demand and Supply

Net Crop Irrigation Requirements

(Depth in mm at the crop)

Crop	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Year
	(millimeters)												
Wheat, Barley & Flax	35	60	95	75	-	-	-	-	-	65	45	50	425
Pulses & Broad Beans	35	60	-	-	-	-	-	-	50	35	45	50	275
Cotton	-	75	35	100	160	260	310	125	-	-	-	-	1,065
Groundnuts	-	-	30	35	100	150	215	200	75	-	-	-	805
Sorghum & Maize	-	-	30	65	200	265	200	-	-	-	-	-	760
Sesame	-	-	-	35	35	150	200	170	30	-	-	-	620
Berseem/Barley Forage	20	40	100	50	-	-	-	65	65	130	50	20	540
Alfalfa Forage	10	40	100	155	250	295	310	265	180	110	70	30	1,665
Citrus	20	35	80	115	170	215	235	235	170	105	40	15	1,435
Vegetables	5	25	50	95	170	230	235	185	160	110	35	10	1,310

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ANNEX 3
Table 2

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Water Demand and Supply

Net Irrigation Requirement for Cropping Patterns
(Depth in mm at the field)

Cropping Pattern	Cropped Area Fraction %	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		millimeters											
Mixed Arable:													
Field crops													
Wheat	50	17	30	47	37	-	-	-	-	-	32	22	25
Flax	20	7	12	19	15	-	-	-	-	-	13	9	10
Pulses, beans	20	7	12	-	-	-	-	-	-	10	7	9	10
Cotton	20	-	15	7	20	32	52	62	25	-	-	-	-
Groundnuts	10	-	-	3	3	10	15	21	20	7	-	-	-
Sesame	10	-	-	-	3	3	15	20	17	3	-	-	-
Subtotal		31	69	76	78	45	82	103	62	20	52	40	45
Forage													
Berseem/barley	100	20	40	100	50	-	-	-	65	65	130	50	20
Sorghum/maize	60	-	-	18	39	120	159	120	-	-	-	-	-
Subtotal		20	40	118	89	120	159	120	65	65	130	50	20
Vegetables	100	5	25	50	95	170	230	235	185	160	110	35	10
Specialist Citrus:													
Citrus	100	20	35	80	115	170	215	235	235	170	105	40	15
Forage													
Alfalfa	50	5	20	50	77	125	147	155	132	90	55	35	15
Berseem/barley	50	10	20	50	25	-	-	-	32	32	65	25	10
Sorghum/maize	15	-	-	4	10	30	40	30	-	-	-	-	-
Subtotal		15	40	104	112	155	187	185	164	122	120	60	25
Vegetables	100	5	25	50	95	170	230	235	185	160	110	35	10
Specialist Vegetables:													
Vegetables	100	5	25	50	95	170	230	235	185	160	110	35	10
Forage (as for Specialist Citrus)		15	40	104	112	155	187	185	164	122	120	60	25

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LOWER KHALIS IRRIGATION PROJECT

Water Demand and Supply

Irrigation Water Requirements

From Gravity and Pumped Sources

Unit	Farm System	Net Area (ha)	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
			(cubic meters per second)												
	Gravity Canal Commands														
	Project Areas:														
I(a) I(b) I(c)	Mixed arable; 62.5% field crops) 31.2% forage) 2/ 6.3% vegetable)	14,700	2.1	4.8	7.3	7.0	6.4	9.6	9.8	6.0	3.5	6.8	3.6	2.9	
II(a)	Mixed arable; 62.5% field crops) 31.2% forage) 2/ 6.3% vegetable)	22,700	3.2	7.4	11.3	10.8	9.9	14.8	15.0	9.2	5.4	10.5	5.6	4.6	
II(b)	Specialist citrus; 67% citrus) 16.5% forage) 2/ 16.5% vegetable)	2,160	0.2	0.4	1.0	1.4	2.1	2.6	2.8	2.6	2.0	1.3	0.5	0.2	
II(c)	Specialist vegetable; 67% vegetable) 33% forage) 2/	8,870	0.5	1.6	3.7	5.2	8.3	10.7	10.8	9.0	7.3	5.9	2.3	0.8	
	Non-project Areas:														
	Orchards	1,170	0.1	0.2	0.5	0.8	1.2	1.5	1.6	1.6	1.2	0.7	0.3	0.1	
	Poultry farm	900	0.2	0.6	0.8	0.7	0.7	1.2	1.4	1.0	0.6	0.7	0.5	0.3	
	Undeveloped 1/	3,000	0.1	0.3	0.5	-	-	-	-	-	-	0.5	-	-	
	Canal losses	-	2.1	2.5	3.8	4.7	5.6	6.5	6.1	5.2	4.6	5.3	2.9	2.4	
	Total gravity	53,500	8.5	17.8	28.9	30.6	34.2	46.9 3/	47.5 3/	34.6	24.6	31.7	15.9	11.5	
	Pump Commands														
	Project Pumps:														
III(a) III(b)	Mixed arable; 62.5% field crops) 25% forage) 2/ 12.5% vegetable)	7,200	1.0	2.3	3.2	3.7	3.6	5.2	5.7	4.2	2.6	3.2	1.8	1.4	
	Canal losses	-	0.1	0.2	0.2	0.3	0.3	0.4	0.4	0.3	0.2	0.3	0.2	0.2	
	Totals	7,200	1.1	2.5	3.4	4.0	3.9	5.6	6.1	4.5	2.8	3.5	2.0	1.6	
	Non-project Pumps:														
	Tigris riverain 2/	1,100	0.2	0.4	0.5	0.6	0.6	0.9	0.9	0.7	0.4	0.5	0.3	0.2	
	Orchards	2,900	0.4	0.6	1.3	1.9	2.8	3.6	3.9	3.9	2.8	1.8	0.7	0.2	
	Canal losses	-	0.1	0.1	0.1	0.2	0.3	0.4	0.4	0.4	0.3	0.2	0.1	-	
	Totals	4,000	0.7	1.1	1.9	2.7	3.7	4.9	5.2	5.0	3.5	2.5	1.1	0.4	
	Total, pumping	11,200	1.8	3.6	5.3	6.7	7.6	10.5	11.3	9.5	6.3	6.0	3.1	2.0	
	Demand		(Millions of cubic meters)												
Demand from Diyala at project canal head, Khalis			22.2	46.5	75.4	79.9	89.3	122.4	124.0	90.3	64.2	82.7	41.5	30.0	868.4
Demand from Tigris at project pumps			2.9	6.5	8.9	10.4	10.2	14.6	15.9	11.7	7.3	9.1	5.2	4.3	107.0
Demand from Tigris at non-project pumps			1.8	2.9	5.0	7.0	9.7	12.8	13.6	13.0	9.1	6.5	2.9	1.0	85.3
Total project area pumping demand from Tigris			4.7	9.4	13.9	17.4	19.9	27.4	29.5	24.7	16.4	15.6	8.1	5.3	192.3
Total project area gravity and pumping demand from Diyala and Tigris			26.9	55.9	89.3	97.3	109.2	149.8	153.5	115.0	80.6	98.3	49.6	35.3	1,060.7

1/ Undeveloped areas assumed to receive 1/3 rate of project supplies during winter.
2/ Areas which would be developed under the project.
3/ Short duration peaks up to 53 m^3/sec anticipated during June and July.

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IRAQ

LOWER KHALIS IRRIGATION PROJECT

Water Demand and Supply

Summary Table

	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	Year
	(millions of m^3)												
Diyala River Discharge above Diyala Weir 1/	140	270	380	340	255	410	475	375	230	355	220	180	3,620
Lower Khalis Project Water Demand from Khalis Canal 2/	22	47	75	80	89	122	124	90	64	83	42	30	868
Lower Diyala Region Water Demand at Partial Development 3/	100	170	280	200	240	360	380	350	230	300	150	160	2,920
Lower Diyala Region Water Demand at Full Development 4/	140	265	380	340	255	405	475	375	225	355	215	180	3,610

1/ 75% probability of flow, with no additional upstream development, in Diyala river as regulated by Derbendi Khan dam.

2/ Gravity supplies, Khalis canal, see Table 4.

3/ Upper and Lower Khalis projects at full development, with remaining areas at present state of development.

4/ At full development, with 15% shortage in summer, of Left and Right Bank of Lower Diyala.

ANNEX 3
Table 5

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IRAQ

LOWER KHALIS IRRIGATION PROJECT

Security of Project Water Supply

1. Most of the project area (85%) would receive its water by gravity from the Diyala river, while about 15% would be supplied by pumping from the Tigris. The gravity supplies would be diverted into the Khalis canal serving the Upper and Lower Khalis areas and which, after remodeling to increase its discharge by 12 m^3/sec, would have adequate capacity to meet the project peak monthly demand of 47 m^3/sec at full development, and short-duration peaks of up to 53 m^3/sec.

2. The Lower Khalis project area constitutes about one-fifth of the total area proposed for irrigation development in the Diyala basin (see Map 1), two-thirds of which would be in the Lower Diyala region, according to the Basin Plan proposed for the Iraqi Government in 1960 by the consultants, Sir M. MacDonald & Partners. The main diversion point for water supplies to the lower part of the basin is the Diyala weir, built in 1942 and reconstructed in 1968 as a gated barrage. The river is controlled by the Derbendi Khan dam, 120 km upstream of the weir. This large rockfill dam has a live storage capacity of 2,500 m^3, and was completed in 1962.

3. About 25% of the catchment area above the reservoir (17,900 km^2) lies in Iraq, and 75% is located in Iran. Of the intermediate catchment between Derbendi Khan and the Diyala weir (12,800 km^2), about 32% lies in Iran. A preliminary study by Bank staff of the areas in the Iranian part of the catchment which may have potential for irrigation from tributaries of the Diyala, indicates that such development would probably be limited to about 28,000 ha. Iran has stated that it intends to develop these areas and in addition to divert flows from the Sirwan, a major tributary of the Diyala, to irrigate 50,000 ha outside the basin, so that reductions in river flows entering Iraq may be expected. The irrigation water requirements per ha for these developments are unlikely to exceed 13,000 m^3/yr estimated on the same basis as for the Lower Diyala region. Full development of the 28,000 ha within the basin would require incremental abstractions by Iran of about 360 Mm^3 per year, or about 7% of the present median flow of the Diyala and 10% of the flow equalled or exceeded in 75% of years at Diyala weir. Diversion to the 50,000 ha area outside the basin might give rise to increases in abstractions of about 750 Mm^3/yr.

4. As part of the review of the security of water supplies for the Lower Khalis project, a dam operation study has been carried out by Bank staff to estimate the frequency and extent of water shortages which might occur if the 1960 Basin Plan is implemented by Iraq, and if all the irrigable areas on the tributaries in Iran are also developed. The Lower

Khalis feasibility study projected that this would occur by 1985, but even if this optimistic target is achieved, water usage is unlikely to reach its peak until 1990.

5. In the Bank study, monthly operations were simulated with inflows derived from 13 years of flow records obtained at Derbendi Khan dam site as well as from 39 years of records taken at Diyala weir, and translated to represent flows at the dam site. Full irrigation development was assumed for the water demand, and releases for hydroelectric power production, if any, were assumed to be of lower priority than irrigation, and were ignored. Irrigation diversions within the basin in Iran were assumed to be 10% of median Diyala flows. The results are shown below.

Simulated Operation of Derbendi Khan Dam at Full Irrigation Development in the Diyala Basin (350,000 ha)

Frequency Years per 100	13-Year Study: Shortage as % of Annual Requirement	39-Year Study: Shortage as % of Annual Requirement
10	42	34
20	26	25
25	17	20
30	9	16
40	0	3
50	0	0

6. The consultants considered that in 95 years out of 100 the water supply should at least equal 90% of the irrigation requirements for optimal plant growth. This is over-cautious and uneconomical, and a more appropriate criterion is that in 75 years out of 100, 90% of the irrigation requirements must be supplied, so that shortages may exceed 10% of the annual requirement in not more than 25 years out of 100.

7. In the 39-year dam operation simulation, the inflow equalled or exceeded in 75% of years would provide for 80% of requirements, or 10% short of the desired norm. There are several ways to achieve the norm including the restriction of irrigation development. The areas to be developed in the basin could be reduced by 10% (35,000 ha), or cropping intensities could be reduced. If, however, all the areas are to be developed, a supplementary source of water supply would have to be provided. Three alternative sources exist:

(a) A 16 m^3/sec standby pump installation on the river Tigris, to discharge supplies into the Khalis canal. It would operate continuously if the dam level is low for the season, and would suffice to increase the level of guarantee to the acceptable range for the basin as a whole, as shown in the table below. This installation would cost about ID 2 million (US$6 million). Shortages might then be expected to occur with the following frequencies, which would be acceptable;

Simulated Operation of Derbendi Khan Dam at Full Irrigation Development in the Diyala Basin with a 16 m^3/sec Standby Pump
39-Year Study

Frequency Years per 100	Shortage as % of Annual Requirements
10	25
20	15
25	10
30	7
40	1
50	0

If incremental abstractions and use by Iran were to be increased to about 1,200 Mm^3/year as a result of diversion outside the basin, an enlargement of the capacity of pumping installation on the Tigris to about 55-60 m^3/sec would meet the Diyala basin requirements. This method of supplementing supplies assumes that sufficient water would be available in the Tigris, in which the ultimate water resource potential is about 33,000 to 38,000 Mm^3 annually. Present minimum flows which occur in October average about 400 m^3/sec near Baghdad. The maximum probable supplementary demand for the Lower Khalis project would not be more than 870 Mm^3/yr and 47 m^3/sec, and 330 m^3 and about 8-14 m^3/sec for the Upper Khalis area, and would be fairly small in comparison with flows presently available in the Tigris.

(b) Another possible method would be by means of the Lesser Zab-Diyala feeder canal. This scheme is only worth consideration if the Kirkuk irrigation project is constructed. Even then, since the Diyala basin would be on the tail end of this system, supplies might be unreliable.

(c) A third possibility would be to build the Jebel Hamrin dam at a site a short distance upstream of the Diyala weir. By regulating the intermediate flow it would provide another 500-700 Mm^3 annually, enough to raise the security of the water supply to acceptable levels if no diversions are made to developments outside the basin. Although this alternative has not been studied in detail, at an estimated cost of about ID 7 million (US$21 million) it would probably not compare favorably with the pump scheme. If, however, hydroelectric turbines are to be installed at Derbendi Khan, the Jebel Hamrin dam could also be used for re-regulation of discharge. If the latter is not built, demand for firm power will cause water to be wasted at times of low irrigation demands. In this case the Jebel Hamrin dam might be the most economical supplementary supply.

8. The present and projected water supply and demand for the Lower Diyala region, assuming future increase in the use of Diyala headwaters, are shown in Table 1, together with the possible pumping demand from the Tigris to provide for shortages.

9. After the study referred to in para 3 had been completed, the Iranian government transmitted to the Bank it's new development plan for that part of Diyala basin lying within its territory. The plan was prepared by Consulting Engineers, MAHAB, and indicates the possibility of future water use from the Diyala tributaries up to the order of 2,000 Mm^3/year within the next 15 years. This would compare with 1,200 Mm^3/year previously anticipated (para 7). If 2,000 Mm^3 depletion were to take place, a supplementary supply to meet the project demand could be obtained by expanding the pumping capacity on the Tigris from the 55-60 m^3/sec proposed in para 7 to about 90 m^3/sec. The flow of the Tigris is more than adequate to meet the expanded demand, since even the minimum flow of the Tigris is about 400 m^3/sec near Baghdad. Therefore, the run-off depletion of the Diyala due to future water use in Iranian territory, would not affect the viability of the Lower Khalis Project.

October 31, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Security of Project Water Supply
Water Supply and Demand for Lower Diyala Region
(with provision for possible future increase in use of Diyala headwaters)

	Annual Flow/Demand			Critical Month Flow/Demand		
	Present	Future	Increment	Present	Future	Month
	------- million m^3 -------			---- m^3/sec ----		
A. DIYALA RIVER						
Flow at Diyala Weir in 75% of years	3,620	3,620	-	54	54	Jan 1/
Demand, Lower Diyala Region						
Lower Khalis Project	360	870	510	6	9	Jan
Upper Khalis Project	500	600	100	7	8	Jan
Other Projects	1,510	2,550	1,040	24	37	Jan
Subtotal, Optimum	2,370	4,020	1,650	37	54	Jan
Acceptable Shortage 2/	-	400	400	-	-	
Withdrawal at Diyala Weir	2,370	3,620	1,250	125	185	July 3/
Surplus at Diyala Weir	1,250	0	-	60	0	
Possible Future Increase in Use of Diyala Headwaters	-	1,150	1,150	-	-	
Deficit at Diyala Weir	-	1,150	1,150	-	-	
B. TIGRIS RIVER						
Mean Flow at Baghdad	38,000	32,900	0	403	403	Oct 4/
Possible Future Demand for Lower Diyala Region						
Lower Khalis Project Pumps	225	190	-35	20	6	Oct
Pumping for Substitution for Diyala Water:						
- to Lower Khalis	-	870	870	-	32	Oct
- to Upper Khalis	-	330	330	-	10	Oct
	225	1,390	1,165		48	Oct
(% of future flow of Tigris at Baghdad)			(3%)		(12%)	

1/ Critical because of conflicting requirements between irrigation and power.
2/ Design provides for 90% of full annual demand in 75% of years, with all of 10% shortage distributed to summer crops.
3/ Critical because of maximum demand with acceptable shortage.
4/ Critical because of low flow in Tigris.

June 22, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Project Works

Introduction

1. The Lower Khalis Project is part of the Diyala Irrigation Project for which a development program has been operative since the late 1950's. Lower Khalis is planned to be a rapid concentrated program of investment, with a more diverse agricultural plan, more intensive farming systems and correspondingly higher productivity than in the other components of the Diyala basin development.

2. Engineering works already constructed and which affect the project, are shown in Maps 1 and 2 and are as follows:

(a) Derbendi Khan Dam: This large rock fill dam, which was completed in 1962 at a cost of ID 20 million, is situated on the Diyala river and controls 55% of the total basin catchment. The reservoir has a live storage capacity of 2,500 Mm^3.

(b) Diyala Weir: This is a fully controlled barrage across the Diyala river at the head of the Lower Diyala plain, with right and left bank feeder canal head regulators. It was substantially completed in 1969 at a cost of ID 1.8 million.

(c) Khalis Main Canal: This is the right bank feeder canal taking off at Diyala weir. The first 52.5 km from the weir to Khalis town were completed in 1971 at a total cost of ID 1.0 million. It has at present a full supply discharge capacity of 75 m^3/sec at the head, and 41 m^3/ sec at the tail for the Lower Khalis Project.

(d) Upper Khalis Project: The irrigation distribution and main drainage systems have been largely reconstructed in the northern area served by the Khalis main canal, throughout a net commanded of 57,700 ha. The work will be completed in 1972, at a cost of about ID 3 million (excluding the cost of the main canal reconstruction). Collector drains will be constructed under the next stage of the Upper Khalis Project.

(e) Daudiya Outfall Drain and Pump Station: These were completed in 1963, at a total cost of ID 0.4 million; the drain has a total length of 28 km, and links with Khalis

outfall drain at Dokhela. The drain has a minimum depth of 3 m below ground level. The pump station discharges into the Tigris, with a design capacity 8 m^3/sec at 2.5 m head.

(f) Roads: In addition to the old Baghdad - Baquba road via Bani Saad, the new Baghdad - Kirkuk road passes through the project area for a total distance of 50 km. This is a first class surfaced road and was completed in 1969.

3. The existing rudimentary irrigation system in the Lower Khalis area is supplied from three branch canals which have a combined capacity of only about 12 m^3/sec. These branch canals are fed from the Khalis main canal, of which the original tail reach has not been reconstructed under the Upper Khalis Project works (para 2(c) above), and which has an existing capacity of about 11 m^3/sec. Irrigated cropping intensities of only about 40% in winter and 10% in summer can presently be supported on about 37,400 ha of the project area. Channel slopes, especially in the tail reaches, are very shallow and this leads to heavy siltation particularly in spring. As a result, a number of distributary canals have either gone out of use or have to be served by pumps. There are virtually no water control structures other than simple offtakes to the existing haphazard system of watercourses, and canal alignments are tortuous and often duplicated, with inadequate capacity and water level command over the area served.

Project Irrigation Supply and Distribution System

4. The project has been divided into three Development Areas, I, II and III, as shown in Map 2, which form a basis for the program of implementation and project investment. The existing irrigation system in the project area would be replaced by a new system of unlined canals which would supply the project from two sources:

(a) by gravity from the Khalis main canal, to serve Development Areas I and II (48,400 ha). This area includes lands presently served by Diyala pumps which would become redundant; and

(b) by pumping from the Tigris river to serve Development Area III (8,300 ha).

The new system is illustrated in Map 2. To supply the gravity system at full development, the capacity of the existing 53-km length Khalis main canal might have to be increased by 12 m^3/sec, from about 80 to 92 m^3/sec at its head and from 41 to 53 m^3/sec at its tail (which is at the head of the project area), unless a supplementary supply is to be provided by pumping from the Tigris near Khalis (see Annex 4). The head capacity of 53 m^3/sec required to meet short-duration summer peaks, represents a gross project water requirement, including losses of 1.02 litres/sec/ha.

5. The Khalis main canal would be extended for 48 km into the project area where it would serve a network of 11 branch and 75 distributary canals. For the lands to the west, adjacent to the Tigris river, which cannot be

commanded by the gravity system, two new pumping stations, Rashidiyah North (2.1 m^3/sec) and South (3.0 m^3/sec) would be constructed. Some existing pumping stations in this area would continue to supply the riverain strip (1,100 ha) while others would serve established orchards, or become redundant. The new pumping stations would serve a network of six distributary canals. The total length of branch and distributary canals in the project is 538 km. Each distributary would supply a number of watercourses serving blocks of land (watercourse units) varying in size from 48-80 ha, depending upon the cropping pattern (Annex 10). There would be about 850 such watercourse units. The watercourse is the smallest channel constructed by the project; on-farm channels would be constructed by the cultivators in the course of farming operations. Watercourse unit areas and designed water use and outlet discharges would be as follows:

Cropping System	Watercourse Unit Area (ha)	Design Use Rate (l/sec/ha)	Outlet Discharge (l/sec)
Mixed Arable	75-80	0.62-0.66	47-49
Citrus	48	0.95	46
Vegetable	60	0.72-0.80	43-48

6. Detailed consideration has been given to lining of canals and to transit losses in unlined canals. It has been concluded that unlined earth canals would be more economical than lined canals and would be suitable for the project, with an allowance for transit losses equivalent to 2.5 m^3/sec/million m^2 of wetted channel perimeter. The canals have been designed as regime channels using the Lacey theory, to transport sediment through the system without scouring.

7. The canals would be fully controlled by manually-operated regulators, including cross regulators and head regulators at canal offtakes, and cross regulators in distributary canals at outlets to watercourses. To enable simple direct measurement of flows to be made, all except the largest regulators would be provided with movable broad-crested weirs instead of the conventional undershot gates.

8. The designs and types of construction proposed in the feasibility report have been fully developed on other irrigation schemes in Iraq and elsewhere. Similar major works are nearing completion in the Lower Diyala region, and very satisfactory engineering construction standards are being achieved.

9. The distribution system would be controlable down to, and including, adjustable watercourse outlets, so that farmers would receive supplies reasonably on demand. The smallest structures to be built by the project would be the farm outlets from watercourses. To assist in the control and regulation of the system, an escape would be provided to the Diyala River from the Khalis main canal extension as well as escapes to drains at the tail of every distributary. Bridges or culverts would be provided at all road crossings.

On-farm Drainage Works

10. Need for Drainage. Under existing conditions, watertable levels of 1.5 to about 5 m occur with irrigation, rising by about 1 m seasonally, causing salinity concentration in the root zone and tending to produce waterlogging. It is probable that under future project conditions there would be an overall rise in the watertable of about 1 m per year unless drainage is provided. To control the depth of groundwater with increased irrigation supplies and to allow leaching of salts, subsoil drainage would be required throughout the project area within a year or two of the start of more intensive irrigation development.

11. Type of Drainage. Drainage and control of groundwater can be obtained by using either vertical tubewells or horizontal buried field drains. The feasibility of tubewells would depend on the presence of suitably permeable upper soils and a deep aquifer. From the available information from wells and deep borings, no extensive aquifers of adequate depth exist in the project area. The small aquifers which do occur would require the use of many small tubewells, which would be more expensive than the alternative buried field drains. A further disadvantage of the use of tubewells would be the unacceptably large quantities of dissolved salts which would have to be pumped into the rivers for many years before the existing highly saline groundwater could be replaced by fresher water. Drainage by an open channel system supplemented by buried field drains would therefore be adopted for the project.

12. Design. An analysis by the consultants for the feasibility report (MacDonalds) indicates that the optimum depth and spacing of open collector drains would be 2.5 m and 600 m respectively, and would provide a depth of buried field drains of 2.2 m at their outlets. The spacing of field drains would depend on the permeability of the soils, on the transmission characteristics of the aquifer and on the results of field trials of the buried drains under future average flow conditions. Three drainage trial areas are being established in the Upper Khalis area, north of the project boundary. Measurements elsewhere in Iraq have indicated that average soil drainage transmissibility values of between 10 and 40 m^3/meter of aquifer width/day (m^2/day) occur. For present purposes, transmissibility is conservatively estimated at 10 m^2/day in most of the area. Low transmissibilities have been observed in the basin depression areas, and for soils in their vicinity an average transmissibility of 5 m^2/day has been assumed. The buried field drain spacings required as a result of these assumptions are estimated to be as follows:

For mixed arable farms on land classified as satisfactory; 200 m.

For mixed arable farms on land classified as questionable; 100 m.

For citrus and vegetable specialist farms; 150 m.

These spacings compare with an average spacing of 55 m for all mixed arable land, and 90 m for specialist farms included in the feasibility report, which are considered to be excessively conservative estimates of drain spacing.

13. The field drains would be 300 m long and between 1.9 and 2.2 m deep, and would discharge into open collector drains. The drainpipes would be continuous, slotted corrugated plastic pipe 8 and 10 cm in diameter, surrounded by a gravel filter laid in a trench excavated by machine. The plastic pipe would be less expensive than locally-made clay or concrete tiles, lighter in weight, more speedily installed and less liable to placing errors. Imported pipe of this type is at present being installed at a sugar estate in southern Iraq, and some lengths will be installed in the Upper Khalis drainage trial areas. Particular attention would have to be paid to the frequency and maximum practical length for cleaning silt from a buried drain; if the cleaning equipment can only operate in lengths of less than 300 m, it may be necessary to install manholes.

Drainage Disposal System

14. An open channel drainage system with a layout and capacity complementary to the irrigation system and totalling 1,450 km would be constructed. It would dispose of water from the buried field drain system, and emergency escape water from the tails of distributary canals. Surface runoff, from rainfall or irrigation excess, would be prevented from entering the open drains directly. The maximum anticipated outflow from the project is 19 m^3/sec, 0.31 litres/sec/ha.

15. Open collector drains have been selected in preference to closed pipe collectors, despite the consequent reduction in cropped land, since the cost would be the same and they would require a shallower main outfall system. Inspection and maintenance, both of buried field drains and collectors, would be simplified with a system of open collectors. Further practical advantages are that open collectors would facilitate land redistribution procedures as they would effectively delineate farm areas, and would enforce more efficient water distribution earlier in the project, as the existing irrigation system would be intersected by the drains and become useless.

16. The drainage system would consist of four separate networks, each comprising an outfall drain (one, Daudiya Outfall, was constructed in 1963), main collector drains and collector drains (600 m apart). The depth of drains would normally range between 2.5 and 3.5 m. The two smaller drainage systems, KN18 and Rashad, would discharge into the Diyala river by gravity; only during exceptional floods would this not be possible, and no pumping stations are necessary. Most of the project drainage would discharge into the Tigris River through the Daudiya and Khalis South systems. A pumping station of 8 m^3/sec capacity would be constructed at the outfall of the Khalis South system at Safia, and the existing Daudiya drainage pumping station would eventually be enlarged in 1980-82 to increase its capacity from 8 to 10 m^3/sec. The pumping stations would be used during the period when high river levels prevent outflow by gravity (usually December to July); at other times they would be bypassed.

Land Levelling

17. Efficient land levelling would be essential for satisfactory water distribution on fields and for the general reduction and uniform control of soil salinity. Experience in Iraq indicates that levelling by farmers, or by project forces on request by farmers, during the development period of a project proceeds very slowly and results in inefficiency. Land levelling would therefore be executed as part of the project works, and would consist of filling in abandoned watercourses and field ditches, basic levelling by scrapers and land planing after ploughing. The levelling would be carried out in each watercourse unit throughout practically all the land to be developed under the project.

18. From surveys and sample areas, an average earth movement of about 230 m^3/ha would be expected in the project area, at a cost of about ID 30 per ha. Land planing to maintain the levelled areas would be required at intervals subsequently.

Roads

19. Although the project area is served with sufficient first class asphalt roads, there is a definite need for second class all-weather roads to open up the central part of the area and enable the speedy movement of machinery and trucks. Roads are planned parallel to two of the branch canals, along all the river embankments, the Baghdad bund and along one of the distributaries. It is also planned to connect all existing major villages by gravel roads to the main road system. The 180 km of gravel roads will be 6.00 meters wide with a gravel thickness of 0.15 meters and be established on earth embankments.

20. Inspection and maintenance roads would be provided along all canal and drain embankments. Their use should be confined to official operation and maintenance traffic. They are not suitable or intended for public traffic, being vulnerable to damage and dangerous, particularly after rainfall. A separate system of farm access roads on earth embankments parallel and adjacent to irrigation channels would be provided.

Flood Protection

21. Derbendi Khan dam has been designed to limit normal discharges below Diyala weir to less than 3,000 m^3/sec. In abnormal circumstances however the discharge below the weir may be higher, but the return period of such an occurrence is 200 years.

22. Future construction of a further dam, the Hamrin dam, mainly for flood purposes, 10 kilometres upstream from Diyala weir, is at present under consideration. The construction of this dam would increase the minimum recurrence interval of the flood required to produce a flow of greater than 3,000 m^3/sec at Diyala weir to 3,000 years.

23. For either of these schemes to be fully effective, the river Diyala channel below Diyala weir must have a capacity of 3,000 m^3/sec. A survey has been carried out which indicates that its present capacity is 2,000 m^3/sec. However, by extending and raising the river embankments below Baquba, this could be increased to the required capacity. It is therefore planned under the Lower Khalis Project to raise the right embankment of the river to a height such that a discharge of 3,000 m^3/sec can be passed safely. This will ensure a satisfactory degree of protection to the project area from Diyala flood flows.

Project Administration Facilities

24. Office Accommodation. Office accommodation would be built for both headquarters and site staff at the main Administration headquarters in Baquba and at the Lower Khalis divisional headquarters at Bani Saad. At Khalis, Bani Saad, Waziriya and Daudiya sub divisional offices will be set up. About 70 small site offices each with office and housing, are proposed for agricultural block offices and canal regulator control points.

25. Housing. A number of houses have been constructed at Baquba and Daudiya under previous contracts for the construction of the Upper Khalis irrigation and drainage project and the Daudiya pump station. These houses will become available, but further housing and facilities would be required for both the Authority and Lower Khalis project staff. Housing would also be required for construction supervision staff.

26. Laboratory. A central Lower Khalis laboratory will be set up at Bani Saad. This laboratory would handle all materials testing in connection with the project construction contracts, and the soil and water testing requirements of the agricultural service.

27. Workshop. For routine maintenance on both plant and mechanical equipment a project workshop will be established at Bani Saad and equipped with all the necessary tools. A project spare parts and general store would be incorporated in the workshop.

28. Telephone System. A project telephone system will be established to link all offices and irrigation control points. This system will be linked up with a similar one presently planned for Upper Khalis.

Construction Contracts

29. The following contracts would be awarded for carrying out the project civil works:

Contract No.	Description	Estimated Start of Contract
K.1	Irrigation & Drainage System, Development Area I	1st quarter, 1975
K.2	Irrigation & Drainage System, Development Area II	3rd quarter, 1974
K.3	Irrigation & Drainage System, Development Area III	2nd quarter, 1974
K.4	DBA Offices, facilities and Housing	4th quarter, 1973

30. Contracts for the comprehensive engineering development of existing irrigation areas, in which water supplies frequently have to be continued during construction, are difficult to phase and carry out. Bidders should have had experience of, and allow for, the problems if they are to submit appropriate prices. The Government is aware of the delaying effect and increased supervision staff requirements if a firm which has inadequate experience, financial resources, equipment and staff is awarded a contract of this type, as happened in the Upper Khalis Project. For this reason, particular care would have to be taken in the evaluation and acceptance of bidders.

31. With the increasing rate of development in Iraq, shortages of skilled tradesmen and supervisors will occur, and restrictions on the employment of foreign staff should be relaxed so far as possible if delays due to insufficient staff are to be avoided. The Iraqi labor laws tend to inhibit overtime work and the retrenchment of contractors' labor forces. This increases costs and construction time, and the laws might with advantage be modified for major civil engineering works in the interests of developments such as the project.

32. Contractors may have difficulty in obtaining imported materials of the required standards in a timely manner. The importation of construction equipment and transport vehicles required by contractors may also be subject to delays. Delays could be prevented if present procedures for issuing import licences would be relaxed and expedited for items listed by contractors in their bids.

33. To prevent delays and possible claims by contractors, it would be important to settle all matters relating to land reallocation, compensation, "freezing" of cultivation, and access, before construction is started in each sub-area. The construction and commissioning program would be linked with the "freezing" of each sub-area throughout one farming season, and this program should if possible be indicated in bid documents so as to reduce uncertainties and bid prices. Table 1 of Annex 6 shows the presently estimated rate of commissioning of development units.

34. Contractors should be enabled to claim for additional costs due either to escalation of prices and wages according to a formula to be specified in the bid documents, or to Government taxes or levies not announced or applicable at the bid opening date.

35. On completion of construction of each sub-area, including on-farm works, it would be inspected by the supervisory staff, accepted if satisfactory, and handed over to the Administration. A completion certificate would be issued to the contractor, who would thereafter be responsible for making good any defects of workmanship during the following year. The engineering section of the Divisional Services Office of the Administration would be responsible for routine maintenance and operation of each sub-area after it is handed over.

October 19, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Project Works

Summary of Project Engineering Works

Description	Estimated Quantities			
	Development Area			Total
	I	II	III	
Irrigation and Drainage Works				
Main Canal:				
New length (km)	-	48	-	48
Earthworks (Mm^3)	-	53	-	53
Cross regulators	-	6	-	6
Syphons	-	6	-	6
Branch Canals:				
Length (km)	59	47	-	106
Earthworks(Mm^3)	1.4	0.7	-	2.1
Head regulators	5	6	-	11
Cross regulators	9	6	-	15
Other structures	5	2	-	7
Distributary Canals:				
Length (km)	138	258	36	432
Earthworks (Mm^3)	1.5	2.7	0.4	4.6
Head regulators	34	44	1	79
Distributary outlets	140	273	46	459
Tail escapes	30	44	7	81
Pump stations	-	-	2	2
Other structures	7	6	2	15
Watercourses:				
Length (km)	875	1685	301	2861
Culverts	362	654	126	1088
Outfall Drains:				
Length (km)	-	82	-	82
Earthworks (Mm^3)	-	1.7	-	1.7
Pump stations	-	1	-	1
Outfalls	-	1	-	1
Box culverts	-	12	-	12

continued...

Description	Estimated Quantities			
	Development Area			Total
	I	II	III	
Main Collector Drains:				
Length (km)	159	157	40	356
Earthworks (Mm^3)	3.0	2.9	0.8	6.7
Box culverts	14	14	8	36
Pipe culverts	328	335	148	811
Underpasses	6	4	2	12
Falls	6	7	2	15
Collector Drains:				
Length (km)	329	607	173	1109
Earthworks (Mm^3)	4.5	8.3	2.4	15.2
Pipe culverts	16	162	36	214
Diyala River Floodbank:				
Earthworks (Mm^3)	-	3.0	-	3.0
Roadworks:				
Length (km)	36	122	24	182
Field Drains:				
Length (km)/1	1100	2300	400	3800
Land Levelling:				
Earthquakes (Mm^3)/1	3.7	7.6	1.7	13.0

Total Earthwork and Concrete Quantities

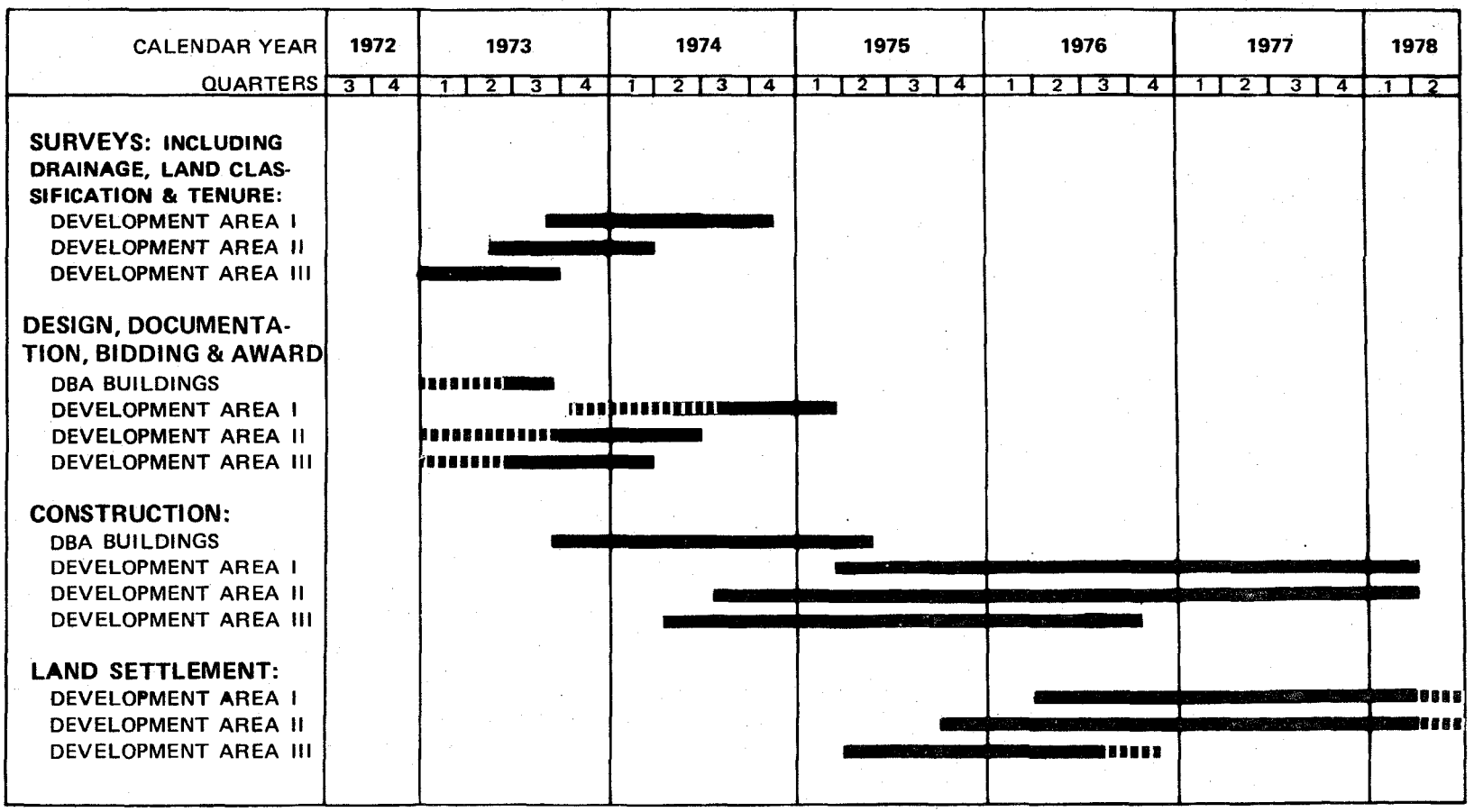
Canal earthwork	8.7 Mm^3
Drain earthwork	23.6 Mm^3
Flood bund earthwork	3.0 Mm^3
Concrete	40,000 m^3

/1 First approximation; to be determined by preconstruction survey.

April 4, 1972

IRAQ
LOWER KHALIS IRRIGATION PROJECT

CONSTRUCTION SCHEDULE



ANNEX 6

World Bank-6691(2R)

ANNEX 6
Table 1

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Construction Schedule

Projected Phasing of Commissioning of Development Units

Commissioning Phase:-		(1)	(2)	(3)	(4)	(5)	(6)	Total
Date:-		Oct '75	Mar '76	Oct '76	Mar '77	Oct '77	Mar '78	
Development Unit	Contract No.	Project net cultivable area, ha						
III(a)	K.3	2,700	-	-	-	-	-	2,700
III(b)	K.3		5,600	-	-	-	-	5,600
II(b)	K.2		2,150	-	-	-	-	2,150
II(c)	K.2		5,250	-	3,600	-	-	8,850
II(a)	K.2			6,500	3,150	7,500	5,550	22,700
I(a)(b)(c)	K.1			3,750	4,250	4,000	2,700	14,700
Totals		2,700	13,000	10,250	11,000	11,500	8,250	56,700

June 22, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Cost Estimates

	Facilities & Administration	Development Areas			Project Costs		
		Area I	Area II	Area III	Local	Foreign	Total
	(ID millions)						
Irrigation Distribution System:							
Main canal	-	-	0.60	-	0.21	0.39	0.60
Branch & distributary canals	-	0.57	0.78	0.29	0.94	0.70	1.64
Watercourses	-	0.14	0.30	0.07	0.28	0.23	0.51
Subtotal					1.43	1.32	2.75
Drainage Disposal System:							
Outfall, main & collector drains & pump station	-	1.73	3.19	0.66	2.55	3.03	5.58
On-farm Development:							
Land levelling	-	0.48	1.00	0.23	0.68	1.03	1.71
Field drainage	-	0.68	1.42	0.28	0.83	1.55	2.38
Farm housing	0.20	-	-	-	0.20	-	0.20
Subtotal					1.71	2.58	4.29
Roads:							
Village roads	-	0.24	0.84	0.17	0.85	0.40	1.25
Ancillary Works:							
Flood protection bund	-	-	0.43	-	0.19	0.24	0.43
Buildings:							
Lower Khalis Division	-	0.15	0.24	0.20	0.59	-	0.59
Headquarters & Upper Khalis	0.68	-	-	-	0.51	0.17	0.68
General Items:	0.02	0.13	0.25	0.06	0.31	0.15	0.46
Subtotal					1.70	0.46	2.16
Total, Contract Works (excl. contingencies)	0.90	4.12	9.05	1.96	8.14	7.89	16.03
Administration:							
Headquarters	0.39	-	-	-	0.39	-	0.39
Lower Khalis Division	0.10	-	-	-	0.10	-	0.10
Subtotal	0.49				0.49		0.49
Consultants:							
Engineering supervision	1.70	-	-	-	0.60	1.10	1.70
Technical services	0.66	-	-	-	0.22	0.44	0.66
Subtotal	2.36				0.82	1.54	2.36
Maintenance Plant:							
Imported equipment	0.80	-	-	-	0.10	0.70	0.80
Land Acquisition:	0.10	-	-	-	0.10	-	0.10
Total (excl. contingencies)	4.65	4.12	9.05	1.96	9.65	10.13	19.78
Contingencies							
Physical (15% of works)	0.13	0.66	1.32	0.29	1.20	1.20	2.40
Prices (5%/yr)	0.87	0.82	1.43	0.30	1.52	1.90	3.42
Subtotal	1.00	1.48	2.75	0.59	2.72	3.10	5.82
Total Project Cost	5.65	5.60	11.80	2.55	12.37	13.23	25.60

June 22, 1972

IRAQ
LOWER KHALIS IRRIGATION PROJECT
Cost Estimates
Estimate of Annual Expenditure

	Fiscal Years (April-March)					
	1972/73	1973/74	1974/75	1975/76	1976/77	1977/78
	(ID millions)					
Irrigation Distribution System:						
Main canal	-	-	0.30	0.30	-	-
Branch and distr'y canals	-	0.05	0.59	0.55	0.29	0.16
Watercourses	-	-	0.17	0.17	0.10	0.07
Drainage Disposal System:						
Outfall, main & collector drains & pump station	-	0.05	2.11	2.00	1.00	0.42
On-farm Development:						
Land levelling	-	-	0.27	0.64	0.53	0.27
Field drainage	-	-	0.39	0.86	0.74	0.39
Farm housing	-	-	0.07	0.07	0.06	-
Roads:						
Village roads	-	-	0.13	0.55	0.44	0.13
Ancillary Works:						
Flood protection bund	-	-	-	0.20	0.13	0.10
Buildings:						
Lower Khalis Division	-	-	0.22	0.20	0.17	-
Headquarters & Upper Khalis	-	0.20	0.48	-	-	-
General Items:	-	0.07	0.13	0.15	0.09	0.02
Total, Contract Works (excl. contingencies)	-	0.37	4.86	5.69	3.55	1.56
Administration:						
Headquarters	0.04	0.04	0.04	0.09	0.10	0.08
Lower Khalis Division	0.02	0.02	0.02	0.02	0.01	0.01
Subtotal	0.06	0.06	0.06	0.11	0.11	0.09
Consultants:						
Engineering supervision	0.05	0.14	0.38	0.51	0.41	0.21
Technical services	0.10	0.12	0.12	0.14	0.13	0.05
Subtotal	0.15	0.26	0.50	0.65	0.54	0.26
Maintenance Plant:						
Imported equipment	-	0.15	0.32	0.17	0.11	0.05
Land Acquisition	0.02	0.08	-	-	-	-
Total, excluding contingencies	0.23	0.92	5.74	6.62	4.31	1.96
Contingencies						
Physical (15% of works)	-	0.06	0.77	0.81	0.54	0.22
Prices (5%/yr)	-	0.17	0.77	1.27	1.07	0.14
Subtotal	-	0.23	1.54	2.08	1.61	0.36
TOTAL	0.23	1.15	7.28	8.70	5.92	2.32

October 17, 1972

ANNEX 7
Table 3

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Cost Estimates

Operation & Maintenance Machinery and Equipment Requirements

Category	Description	Quantity	Estimated Landed Cost (ID 000's) Foreign	Local1/	Total	Foreign (US$ 000's)
A.	Equipment to Be Procured following International Competitive Bidding					
	Hydraulic excavators	10	100	-	100	304
	Motor road graders, 125-150 hp	4	48	-	48	146
	Crawler Tractors, 75-100 hp, with matching 7-cu yd scrapers	2	80	-	80	243
	Crawler tractors, 180-hp, with matching bulldozer equipment	2	50	-	50	152
	Front-end loaders	2	16	-	16	49
	Hydraulic boom ditchers	2	20	-	20	61
	Water tankers	2	6	-	6	18
	Buried field drain cleaning equipment	10	18	-	18	55
	Workshop machinery		30	-	30	91
	Concrete mixers	4	8	-	8	24
	Sump pumps	12	4	-	4	12
	Mobile crane, 5-ton	1	10	-	10	30
	Low loader	1	5	-	5	15
	Fuel and service trucks	2	6	-	6	18
	Trucks, 2 ton	6	18	-	18	55
	Air compressor	1	2	-	2	6
	Sedan cars	16	24	30	54	73
	4-wheel drive vehicles	62	66	59	125	201
	Pick-up trucks	9	27	11	38	82
	Motorcycles	74	19	-	19	58
	Spare parts, 20% for all above items		130	-	130	395
	Total Category A		687	100	787	2,088
B.	Items to Be Procured by Government of Iraq Procedures from Local Agents of Manufacturers					
	Wheeled tractors & miscellaneous small plant		13	-	13	40
	Total Category A & B excl. contingencies		700	100	800	2,128
	Contingencies, 30%		210	30	240	638
	Total including contingencies		910	130	1,040	2,766

1/ Customs duties

June 22, 1972

ANNEX 8

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Estimated Schedule of Disbursements

Exchange Rate: ID 1.00 = US$3.04

Bank Fiscal Year and Quarter	Cumulative Disbursement at End of Quarter (US$ '000)
1972/73	
September 30, 1972	-
December 31, 1972	-
March 31, 1973	100
June 30, 1973	300
1973/74	
September 30, 1973	600
December 31, 1973	900
March 31, 1974	1,300
June 30, 1974	2,500
1974/75	
September 30,1974	3,500
December 31, 1974	4,800
March 31, 1975	6,000
June 30, 1975	7,400
1975/76	
September 30, 1975	10,000
December 30, 1975	12,500
March 31, 1976	15,000
June 30, 1976	17,500
1976/77	
September 30, 1976	20,000
December 31, 1976	22,500
March 31, 1977	25,000
June 30, 1977	27,500
1977/78	
September 30, 1977	30,000
December 31, 1977	33,000
March 31, 1978	36,000
June 30,1978	37,000
1978/79	
September 30, 1978	37,500
December 31, 1978	38,000
March 31, 1979	38,500
June 30, 1979	40,000

October 17, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Organization and Management

General Aspects

1. The Government has set up an Administration to be responsible for project construction and operation in the Upper and Lower Khalis areas, including the area to be developed under the proposed project.

2. A law to establish Agricultural Administrations has been enacted for this purpose and is appended to this Annex. The law and a Presidential decree setting up the project Administration give satisfactory legal powers for management of the project. The Administration would comprise an Executive Board under the chairmanship of a full-time Director General who has been designated and would be responsible to the Minister of Agrarian Reform.

3. The Board would be required to submit their policy recommendations, the plans and programs for development of the project area, an annual budget, progress reports, and bid recommendations for major contracts to the Minister of Agrarian Reform. In addition to approving the project plan and program and annual budgets, and confirmation for the award of major contracts, the Minister or, through him, the Higher Agricultural Council would issue policy, revenue, and other directives for guidance of the Board. Central Government would allocate all necessary funds, and would authorize the Ministry of Finance to provide funds within the approved budget as invoiced by the Board. All revenue collected by the Board would be passed to the Ministry of Finance, which would arrange for audit of the Board's accounts.

4. The Board's duties would be:

(a) to maintain general supervision over planning and implementation of the development plan, and management of the project, in accordance with Government policy and legislation;

(b) to consider all reports, plans and programs submitted by the Director General, and to make the necessary recommendations or approval;

(c) to supervise the disbursement of allocated funds and the collection of revenue;

(d) to direct all land resettlement;

(e) to arrange contracts for the implementation of the project; and

(f) to ensure the necessary cooperation between Government ministries and the Administration.

Activities of the Administration

5. The activities of the Administration would be carried out by the various offices of the organization under the direct control of the office of the Director General. The Director General would be responsible in particular for:

(a) preparation, and submission to the Minister, of reports, plans and programs for development and management of the development of the Upper and Lower Khalis Projects;

(b) implementation of plans and programs as directed by the Board;

(c) supervision of expenditure of funds and revenue collection; and

(d) appointment of staff with full disciplinary control over all staff either employed by or seconded to the Administration.

6. The activities of the Administration would be carried out by the three offices of Project Administration, Project Development and Project Services. Within each office the activities would be grouped into sections as described below.

Project Administrative Office

7. Within the Project Administrative Office, there would be three sections: Personnel, Accounts and Stores. The Personnel Section would be responsible for recruitment and employment. The Accounts Section would be responsible for all Project finances and would also be closely associated with the Stores Section dealing with purchase and administration of all Project stores and equipment. The Accounts Section would also supervise the collection of revenue through the Divisional Offices.

Project Development Office

8. This office would comprise three sections: Planning and Research, Design and Construction, and Land Settlement. The Planning and Research Section would be at the center of the Administration, initiating the activities of the Administration and plans and programs for project development, planning and coordinating services, and carrying out investigations and research. To strengthen this section, the consultants would provide an Engineering Project Manager, a Chief Agriculturalist, and specialists in management, irrigation, drainage, and citrus farming, as soon as possible after the Administration is legally established. Specialists in livestock and farm mechanization would be added later, as indicated in Table 1. This section would be responsible for devising the development programs for the proposed project, outlined in para 11, from which the Administration would derive its

impetus, and which would be prepared by the datelines given in Chapter V of this report. The section would constitute a permanent headquarters technical and research office, but would also be actively engaged in coordination of the various sections of the Divisional Services Office, in field studies, monitoring and research, and liaison with ministries. An important function of this section would be evaluation of technical and economic aspects of the project and the development program would include a detailed plan for sampling to provide the basis for future comparisons.

9. The Design and Construction Section would be concerned mainly with engineering for project development, including designs, specifications, bids, loan applications, supervision of construction, and commissioning. This section is expected to be largely staffed by the project consultants, who would employ such foreign supervisory staff as and when needed to carry out the project works to the required standards and time schedule, since, with the intended rate of development in Iraq, there will be acute shortages of trained and experienced engineers and supervisors in the country. Engineering staff should be sufficient, both in the preparatory period before bid documents for works are issued, and in the finalizing period after substantial completion of contract works, to ensure that neither the start of construction nor the final payments to contractors are delayed.

10. The Land Settlement Section would deal with all land and Agrarian Reform activities associated with project development, including classification, tenure, boundary adjustment, resettlement, distribution and registration. Because of the considerable disturbance to irrigated agriculture which would be unavoidable during project construction, every effort should be made to inform farmers well in advance about the proposed development and the anticipated changes and benefits. This task would be mainly the concern of the Land Settlement Section. Its main work will occur before and during project construction, when temporary arrangements followed by permanent settlement will involved coordination, programming, and planning of social, agricultural and engineering activities. Subsequently it will be required to regulate and negotiate land transfers which will continue to occur and to evaluate production, as well as to determine revenue assessment.

11. The Administration, with the assistance of its consultants, would be required to prepare the following coordinated programs for carrying out the project:

(a) Land Settlement Program: this would include the phasing of collection of data on population, present holdings and their location, mapping and land classification (as obtained from the proposed pre-construction survey); the updating of the Land Register, the preparation of a redistribution plan, and a schedule for publication of new boundaries in relation to the commissioning periods following completion of construction of subareas; required phasing of any new legislation needed for implementation; follow-up activities after resettlement, and the associated staff, housing, office and transport requirements;

(b) Engineering Development Program: this would include the phasing of requirements for air and topographical surveys, drainage and land classification surveys, drainage trials, preparation of designs and bid documents, bidding and evaluation, contract mobilization, construction, commissioning, acceptance and operation of sub-areas, DBA facilities, pump stations, canals, and roads; and the associated staff housing, office and transport requirements;

(c) Agricultural Development and Monitoring Program: this would include the phasing of selection and development of citrus nursery and demonstration farm areas, in-service training courses, survey of project storage, marketing and processing requirements, farming input needs, machinery support and servicing requirements, project monitoring and evaluation including sampling of present conditions and associated staff, housing, office and transport requirements; and

(d) Administrative Support Program: this would include the phasing of the establishment of temporary and permanent office accommodation, stores construction, staff recruitment and equipment procurement.

Project Divisional Services Office

12. The Project Divisional Services Office at Bani Saad would consist of four sections: Engineering, Agriculture, Cooperatives and Revenue, operating through Divisional, Subdivisional, and Block offices.

13. The Agricultural Section would be responsible for the agricultural development within each Division through the Subdivision and Block offices. One of the most important functions of the Section would be the establishment of an efficient and effective extension service maintaining close liaison and contact with the cultivators. In circumstances where traditional conditions are likely to be subject to considerable changes with the introduction of a more intensified system of agriculture, a determined effort must be made to acquaint the farmers with the full implications of proposed project development, and to obtain their cooperation prior to commencement of development. Assistant agricultural extension officers with basic training would be available in numbers well in excess of project requirements, although they would be insufficiently experienced. This section would therefore be strengthened with a Divisional Agriculturalist and experienced agricultural officers to be provided by the project consultants as shown in Table 1.

14. To support the extension service the Development Office would initiate a program of agronomic studies in soil characteristics, crop diseases, pest control, irrigation practices, and the use of fertilizers and insecticides for the project, maximum advantage being taken of the existing training and research organization in Iraq including those of the UNDP. In-service training of staff would be an important function of the Section. A training program would be drawn up in consultation with existing

training establishments, and a demonstration farm for project extension staff training would be established at Baquba.

15. The Agricultural Section would be responsible for the organization of crop and livestock production in accordance with the project's agricultural development plans. The Section would also assess water requirements in conjunction with the Engineering Section, supervise irrigation at Block, farm and field level, and direct reclamation operations. The Section would be concerned with all agronomic practices, and would be closely linked with the Cooperatives Section for the supply, distribution, and use of feed, fertilizer and pesticides, and in the operation of agricultural machinery.

16. The Cooperatives Section would develop, control and supervise cooperatives within the Project, dealing with credit, agricultural supplies, operation and hiring-out of agricultural machinery, marketing and crop processing, in addition to accounting and auditing. The Section should not be directly involved in agricultural extension work, but would coordinate its activities with the Agricultural Section. The Revenue Collection Section would constitute a field organization jointly responsible to the Project Administrative Office, and Land Settlement Section of the Project Development Office.

17. Although the agency responsible for bulk purchase of inputs would be the Directorate General of Cooperatives and Agricultural Production, the project Administration would provide strong technical and managerial support through the Cooperative Officers. It would be responsible for coordinating bulk purchase arrangements, for distribution to the Cooperatives, as well as for supervising the marketing procedures. The project Administration would provide at least one supervisory officer per Cooperative.

18. The Agricultural Bank has agreed to establish a branch in the project area. Applications for credit will be made to the bank with the assistance of the Administration's Agricultural and Cooperative Officers. It is Government policy that this credit should be channeled through Cooperatives, who would also issue the supplies which are the subject of the credit.

Operation and Maintenance

19. The Engineering Section of the Divisional Services Office would be responsible for operation and maintenance of canals and drains after completion of construction (Annex 5, para 35), including all related pumping plant and equipment, for maintenance of the Administration's buildings, and for servicing and maintaining agricultural machinery. In operating the canals, the Engineering Section would liaise with the Agricultural Section through each Block office to confirm irrigation requirements and with the Project Development Office to coordinate supplies. Canals would then be regulated to supply each distributary with the required volume. The division of supplies from each distributary canal would be arranged by the Agricultural Section in conjunction with the cultivators in each Block. The Engineering Section would carry out routine inspection and maintenance of structures, channels and associated roads, and field drains, and would record

daily flows at the head of each distributary canal, and in main and branch canals.

20. All of Development Areas I and II of the project would depend on the Khalis main canal and on the operation of the Diyala weir for the regulation of their supplies of irrigation water. To ensure undivided responsibility for controlling supplies, it would be necessary for the Administration to take over the operation and maintenance of the whole of the main canal in the Upper Khalis Project before construction work started in the gravity-fed areas of the project, i.e. by late 1973. Detailed instructions for the operation and planned routine maintenance of the weir and the Khalis main canal would have to be prepared, with the assistance of the project consultants, by the Administration by that time. Similar instructions for the remaining project works would be required before the various parts of the system are handed over to the Administration by the contractors.

Staff

21. Determination of the staff establishment would be one of the first tasks of the Planning Section of the Project Development Office. Preliminary estimates of staff requirements are given in Table 1. The posts would be graded according to the qualifications required:

Grade	Qualification
I to IV	University degree or equivalent professional qualification
V to VI	Technical college, diploma or equivalent
VII	Trade qualification or school diploma

22. Agricultural extension service technical staff would be allocated on the following basis:

Mixed arable areas	1 per 200 farm units
Citrus specialist areas	1 per 100 farm units
Horticultural specialist areas	1 per 100 farm units

Consultants

23. The aim would be to build up the Administration so that it would be staffed for long-term management by Iraqi personnel. Special requirements during the initial stages of development would be met by the appointment of personnel to be provided by the consultants for specific duties, until such times as local staff had acquired the necessary skills and experience.

Posts recommended to be filled initially by the consultant's personnel are shown in Table 1.

24. Within the Project Development Office, it is envisaged that eight posts in the Planning and Research Section would be temporarily filled by consultants' personnel, assisted by counterpart Iraqi staff under training, who would take over all the duties of the section at a later stage. The duties of the Design and Construction Section would be directly assigned to the Consultants, who would be appointed to undertake the general supervision and administration of construction contracts and technical control of the construction of the project works. This would follow the present Diyala Irrigation Project arrangements, with Iraqi staff in a number of design office and site supervision posts. The Land Settlement Section would be staffed with Iraqi personnel, assisted when necessary by consultants' counterpart staff for temporary special duties.

25. Within the Divisional Services Office, it is envisaged that staff provided by the consultants would be required initially in engineering and agricultural administration posts and also to fill a number of the agricultural officer posts until such time as Iraqi agricultural officers with sufficient experience and in-service training are available.

Staff Appointments

26. To maintain the project development schedule, it would be necessary for the following staff of the Administration's project headquarters to be in post full-time as soon as possible, and preferably within six months of signing of the loan:

Chief Secretary	Chief Agriculturalist
	Planning Engineer
Accountant	Drainage Engineer
Personnel Manager	Soils and Land Classification Specialist
Project Manager	Citrus Specialist
Management Specialist	Settlement Officer

Assistant Settlement Officers

27. With the aim of ensuring that suitable technical staff would be appointed to the Administration, and if provided by the consultants, details of the qualifications and experience of candidates for the following posts should be sent to the Bank for review prior to their appointment:

Project Contracts Manager	Citrus Specialist
Chief Agriculturalist	Marketing Specialist
Soil and Land Classification Specialist	Divisional Engineering Advisor
Drainage Engineer (DBA Headquarters)	Divisional Agricultural Advisor

ANNEX 9
Table 1

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Organization and Management

Preliminary Estimate of Administrations Staff Requirements

Post	No. Required
Project Administrative Office	
Chief Secretary 1/	1
Accountant 1/	1
Personnel Manager 1/	1
Stores Manager	1
Assistant Secretaries	4
Assistant Accountants	4
Clerks	3/
Project Development Office	
Planning and Research Section:	
Project Contracts Manager(Engineer) 1/ 2/	1
Chief Agriculturalist 1/ 2/	1
Management Specialist 1/ 2/	1
Planning Engineer 1/ 2/	1
Irrigation Agronomist	1
Soil and Land Classification Spec. 1/	1
Drainage Engineer 1/ 2/	1
Livestock Specialist 2/	1
Citrus Specialist 1/ 2/	1
Farm Mechanization Specialist 2/	1
Marketing Specialist 2/	1
Chemist	1
Assistant Engineers	3
Draughtsmen	2
Clerks	3/
Design and Construction Section:	As required by Consultants
Land Settlement Section:	
Settlement Officer 1/	1
Assistant Settlement Officers 1/	5 teams
Clerks	3/

Continued...

1/ Required to be in post as soon as possible, before end-1972.
2/ To be provided by project consultants, approximately for period stated.
3/ Numbers as required.

Preliminary Estimate of Administration's Staff Requirements (cont'd)

Post	No. Required
Project Divisional Services Office, Lower Khalis Division	
Engineering Section	
Administration:	
Divisional Engineer	1
Divisional Engineering Advisor 1/	1
Accountant	1
Clerks	2/
Canal Operation:	
Subdivisional Engineer	4
Overseers	2/
Canal Assistants	2/
Gatemen	2/
Canal and Drain Maintenance:	
Subdivisional Engineer	1
Overseers	2/
Plant Operators	2/
Servicing:	
Mechanical Engineer	1
Foremen	2
Fitters	2/
Drivers	2/
Agricultural Section	
Administration:	
Divisional Agriculturalist	1
Divisional Agricultural Advisor 1/	1
Clerks	2/
Arable Crops:	
Assistant Agricultural Advisor 1/	1
Assistant Agricultural Officers	2/
Orchards:	
Assistant Agricultural Advisor 1/	1
Assistant Agricultural Officers	2/

Continued...

1/ To be provided by project consultants for periods stated, subject to review.
2/ Numbers as required.

Preliminary Estimate of Administration's Staff Requirements (cont'd)

Post	No. Required
Agricultural Section (contd.)	
Horticultural Crops:	
Assistant Agricultural Advisor 1/	1
Assistant Agricultural Officers	2/
Livestock:	
Assistant Agricultural Advisor 1/	1
Assistant Agricultural Officers	2
Co-operative Section	
Divisional Cooperative Officer	1
Accountant	1
Clerks	2/
Subdivisional Cooperative Officer	4
Assistant Cooperative Officer	2/
Assistant Cooperative Officer	2/

1/ To be provided by project consultants, subject to review.
2/ Numbers as required

June 22, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Organization and Management

Project Administration
Extracts from the Law

The following is an unofficial translation of Articles one through nine of the Law on Agricultural Administration which was provided to the Bank during loan negotiations. Following passage of this law a Presidential Decree has established the project Administration with jurisdiction over the Upper and Lower Khalis areas.

In the Name of the People
The Revolution Command Council

In accordance with the provisions of paragraph (A) of the Forty-Second Article of the Provisional Constitution and on the basis of the proposal of the Minister of Agrarian Reform which has been approved by the Higher Agricultural Council, the Revolution Command Council, at its meeting held on 30/4/72, decided the promulgation of the following law:

No. (50) of the Year 1972
L A W
Concerning Agricultural Administrations

Article One - For the purpose of this law, the following terms shall have the meanings ascribed to them hereunder:

1. The Council: The Higher Agricultural Council.

2. The Ministry: The Ministry of Agrarian Reform.

3. The Minister: The Minister of Agrarian Reform.

4. The Administration: Any Administration to be established pursuant to this law.

5. The Board of Directors: The Board of Directors of the Administration.

6. The Director-General: The Director-General of the Administration and Chairman of its Board of Directors.

Article Two - 1. The provisions of this Law shall apply to any Administration to be established, and the lands within its jurisdiction determined, by a Republican Decree to be issued upon a proposal by the Minister and the approval of the Council.

2. The Administration shall have juridical personality entitling it to the enjoyment of all the rights and powers necessary for the realization of its objectives. It shall be represented by the Director-General and shall have financial and administrative autonomy so as to enable it to perform its functions within the limits prescribed by this Law.

Article Three - The Administration shall manage and utilize the lands allocated to it and the land placed under its jurisdiction. It shall draw up the agricultural and economic schemes in accordance with the requirements of the development of the national economy and within the framework of the national development plan. For that purpose, it shall, in particular, undertake the following in cooperation with the competent authorities:

1. Reclaiming the lands falling within the domain of its activities and utilizing the land allotted to it.

2. Operating, maintaining and expanding irrigation and drainage networks and effecting additions thereto as may be needed.

3. Determining the method of utilization of the land falling within the domain of its activities and designating the crop rotation therein.

4. Developing and increasing agricultural and livestock resources and processing the produce thereof.

5. Establishing productive and experimental agricultural, livestock, orchards, nurseries and afforestation fields.

6. Recruiting and training the personnel required for the realization of the objectives of the Administration.

7. Constructing and managing the buildings and installations required for the realization of the objectives of the Administration and the dwellings for its employees, constructing modern villages, and providing public services therein.

8. Guiding, developing, encouraging and supporting cooperative agricultural organizations and investors and contributing to (the efforts towards) the raising of economic, health, social, and cultural standards of the inhabitants of the lands falling within the domain of its activities.

9. Marketing the products of the Administration and providing the required storage and transportation facilities; marketing, when necessary, the products of the utilizers in return for a suitable commission.

Article Four - The funds of the Administration shall consist of the following:

1. Appropriations included in the national development plan and in the state general budget.

2. Proceeds of the land of the Administration, rent or any other income derived therefrom.

3. Rent of the machinery, equipment, transport facilities and installations of the Administration.

4. Proceeds of the sale of agricultural, livestock, industrial and other products produced by the Administration.

5. Fees charged by the Administration for services rendered to the investors and to others.

6. Contributions, gifts and bequests.

Article Five - The Administration shall be administered by a Board of Directors. The Board of Directors shall draw up the Administration's agricultural, economic, technical, administrative and financial policy within the general framework of the national development plan and the resolutions of the Higher Agricultural Council. The Board of Directors shall be composed of:

1. The Director-General: Chairman of the Board of Directors.

2. The Deputy Governor (Muhafiz).

3. Four members drawn from among those persons having experience and specialization in agriculture, agricultural economy, irrigation and accountancy.

4. One member representing each of the farming societies, agricultural cooperative societies and the agricultural workers union operating in the land falling within the domain of activities of the Administration.

Article Six - 1. The Director-General and the members of the Board of Directors shall be appointed by a Republican Decree issued upon a proposal by the Minister and the approval of the council. Their appointment shall be for a term of three years renewable.

The Director-General shall possess the necessary experience and specialization. His appointment shall be on a "Special Grade" position.

2. Two reserve members shall be appointed by a Republican Decree upon a proposal by the Minister and the approval of the Council. Consideration shall be given to their experience and specialization (when they are being considered for this appointment).

3. The chairman and members of the Board of Directors shall be paid an annual remuneration to be determined by the Council.

4. The Board of Directors shall meet at least once every month upon the invitation of the chairman. It may also be convened when necessary upon the invitation of the Minister or the Director-General or the majority of the members. The Minister shall preside over the meetings he attends.

5. The Board of Directors has the right to invite any official or person of experience to attend the meetings (of the Board) for the purpose of hearing his views without the right to vote.

6. A majority of the members, including the chairman, shall constitute a quorum for the meeting. Voting shall be by the absolute majority and if votes are equal, the chairman shall have a casting vote.

Article Seven - The Board of Directors shall be responsible for administering the affairs of the Administraiton. It shall pass decisions for the purpose of realizing the objectives of the Administration and, in particular, decisions dealing with the following matters:

1. Preparing the annual schemes for the activities of the Administration and following up their execution.

2. Issuing administrative directives organizing the work or the Administration.

3. Appointing officials and technicians in the salary scale applicable to grade three and above either directly or by means of secondment from other Ministries or institutions.

4. Concluding contracts with Iraqi and foreign consultants, experts and technicians.

5. The execution of works and the purchase of materials and equipment the total cost of each of which does not exceed one hundred thousand Iraqi Dinars in the manner it deems fit, whether through tenders or by a direct invitation or departmentally as the case may be.

6. Approving the budget and the annual report and staffing of the Administration.

7. The transfer of funds within the same article of the budget.

8. Regulating expenditure and the manner of bookkeeping according to the modern methods adopted by similar institutions.

9. The sale of surplus or written off material and equipment in the manner it deems suitable provided the value is charged as Administration's revenue.

10. Writing off lost or damaged goods, the cost of which does not exceed ten thousand Dinars at the time of purchase; and if it exceeds this amount, the goods shall be written off by law.

11. Determine the amount of charges for services rendered to utilizers or to others in accordance with the instructions it issues for this purpose.

12. Delegate to the Director-General or to any member of the Board of Directors the power to exercise some of its faculties or to undertake any task which may serve the objectives of the Administration. The Director-General is empowered to delegate all or part of his powers to the senior officials of the Administration.

Article Eight - The Director-General shall execute the decisions of the Board of Directors immediately upon their issuance save those decisions relating to the questions enumerated herebelow which shall be presented to the Minister for his approval and which shall be considered to have been approved if the Minister does not approve them within fifteen days from the date of their recorded receipt by the Ministry. If, during the said period, the Minister vetoes any decision, the matter shall be referred back to the Board of Directors for consideration de novo. If the latter insists on passing its contested decision, the various viewpoints shall be put to the Council whose decision in this matter shall be final:

1. The annual plan and the annual schemes of activities of the Administration.

2. The annual budget and the annual report on the accounts of the Administration.

3. The cadres of the Administration's officials and employees.

4. Matters relating to borrowing and awarding loans.

5. Contracts and works the cost of each transaction of which exceeds one hundred thousand (ID 100,000) Iraqi Dinars.

Article Nine - 1. The Administration's fiscal year shall begin on the first day of the month of April and shall end on the thirty first day of the month of March of the following year.

2. The accounts of the Administration shall be subject to control by the Office of Financial Control and the control and inspection of the Ministry. This, however, does not exempt the Administration from the task of auditing and inspecting the accounts through its own offices nor from following the rule of auditing before authorizing payments.

The Administration may appoint a chartered accountant to audit its accounts.

3. Assets of the Administration shall be considered public assets and it shall collect debts due to it according to the law for the collection of debts due to the government.

4. The Administration shall be exempt from all duties.

5. If the budget has not been approved before the beginning of the fiscal year, arrangements shall be made according to the previous or the proposed budget, whichever is less, on the basis of one twelfth of the lesser appropriations until the budget is approved.

June 26, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Agricultural Production

Farm Pattern

1. Under present conditions most farms are under 10 ha and are mainly devoted to arable cropping with an open fallow rotation system, resulting in a low intensity. The main winter crops are barley, wheat, linseed and pulses, while some sesame and cotton are grown in summer. Vegetables and citrus are grown on the river levees. Animal husbandry is mainly in sheep and goat raising.

2. The provision of drainage, irrigation and other inputs under the project would bring a highly systematic and intensive agriculture into the project area which is currently subject to low intensity and yields with practically no prospect for improvement without the project. This would require a number of structural changes in the farming system. The project farming system would be reorganized into the farm types and locational distribution shown in Table 1:

Table 1: FARM TYPES AND CROP AREA

Farm type	Development Unit	No. of Farms	Area Under Each Crop					
			Field Crops	Specialist Vegetables	Non-specialist Vegetables	Citrus	Forage	Total Area
			(hectares)					
8-ha Mixed Arable	I(a,b,c)	4,670	5.0	-	0.5	-	2.5	37,370
4-ha Mixed Arable	III(a,b)	2,090	2.5	-	0.5	-	1.0	8,350
3-ha Citrus	II(b)	720	-	0.5	-	2.0	0.5	2,160
3.6 ha Vegetable	II(c)	2,440	-	2.0	-	-	1.6	8,870
Total		9,920	28,500	5,240	3,380	1,440	18,110	56,750

3. The largest part of the project area, covering some 45,700 ha, would consist of 8-ha and 4-ha mixed arable farming systems of arable field crops, forage and vegetables. The 8-ha farms are projected in areas with relatively low population density and the 4-ha farms in areas of high population density. Vegetable and citrus areas would be located in the horticultural sections of Development Area II because of its soil fertility, proximity to Baghdad and the projected population density.

4. The mixed arable farms would include field crop rotations on about 60% of their area, consisting of the winter crops of wheat pulses, broad beans and linseed and the summer crops of cotton, groundnuts, sesame and maize. All these crops, except groundnuts, are familiar to the local farmers. Forage lands of mixed arable farms, as well as on citrus and vegetable farms, are expected to concentrate mainly on sheep rearing with substantially higher productivity compared to the current sheep industry, resulting from improved nutrition and management. Vegetables would consist of some 16 items, but tomatoes, potatoes and brassica would have the largest output.

5. The citrus specialist farms would have 0.5 ha forage and 0.5 ha vegetables. In addition, the young citrus plantations would also be intercropped with vegetables. This diversification of citrus farms would bring in cash income during the early years of the citrus development period as well as realizing an even seasonal family labor utilization thereafter. The specialist vegetable farms would concentrate on the intensive cultivation of vegetables as a main cash crop on 2 ha and are expected to achieve a 20% higher output of vegetables per hectare than the mixed farms whose primary income would come from arable crops. The remaining effort of the specialist vegetable farms would be devoted to the 1.6 ha of forage.

Cropping Intensities

6. The present cropping intensity on the arable lands is about 40-45% in winter and 10% in summer. Under the project irrigation system the intensities would gradually increase following desalinization, improved irrigation practices and farmers' experience. Cropping intensities of the arable crops are projected to increase to 130%, of which winter and summer intensities would be 90% and 40%, respectively, from the base year cropping intensities of 10% in summer and 66% in winter. The following tables show the projected changes in cropping patterns and intensities for the arable area. Note that barley is expected to disappear after five years while wheat would remain constant from the fourth year.

Table 2: FUTURE CROPPING PATTERNS EXPRESSED AS PERCENTAGE OF ARABLE AREA

Year:	1	2	3	4	5	10
Winter Crops:						
Wheat	20	30	40	50	50	50
Barley	40	30	20	10	5	-
Linseed	2	2	2	5	10	20
Broad Beans	2	2	2	5	5	10
Winter Pulses	2	2	2	5	5	10
Summer Crops:						
Cotton	5	5	5	10	15	20
Sesame	3	3	4	5	5	10
Groundnuts	1	2	2	5	5	5
Maize	1	1	1	2	2	5
Summer/Winter Intensity	10/66	11/66	12/66	22/75	27/75	40/90
Total Intensity	76	77	78	97	102	130

Table 3: SUMMER/WINTER CROPPING INTENSITIES FOR VEGETABLES, CITRUS AND FODDER CROPS

Year:	1	5	10	15
Crop				
Specialist Vegetables	40/50	50/60	65/70	70/75
Non-specialist Vegetables	32/40	40/48	51/56	56/60
Citrus	33/33	100/100	100/100	100/100
Fodder for Sheep	60/100	60/100	60/100	60/100

Yield Projections

7. Present and future yield and total project area under each crop and total output at full development are summarized in Table 4. First-year yields of arable crops are assumed to be slightly higher (about 20%) than the present yields on account of the effect of leaching. The full development yields are expected to be 200-300% higher, whereas the annual yield increase for these crops without the project are assumed to be less than 1% (0.08%).

Table 4: CROP YIELD PROJECTIONS

Crop	Present	Base Year 1	Year 5	Year 10	Year 15	Cropped Area at Full Development	Total Output '000 tons
	(ton/ha)						
Wheat	1.0	1.3	1.8	2.5	3.0	14,300	42.9
Barley	1.3	1.5	1.5	-	-	-	-
Linseed	0.7	0.8	1.0	1.2	1.5	5,700	8.6
Broad Beans	0.8	1.0	1.25	1.5	2.0	2,850	5.7
Winter Pulses	0.7	0.8	0.9	1.0	1.2	2,850	3.4
Sesame	0.4	0.45	0.6	0.8	1.0	2,850	2.9
Maize	1.0	1.15	1.7	2.5	3.0	1,400	4.2
Groundnuts	0.8	0.9	1.25	1.7	2.0	1,400	2.8
Cotton	0.8	1.0	1.6	2.4	3.0	5,700	17.1
Citrus	-	-	4.0	20.0	25.0	1,440	36.0
Berseem-Barley	-	30.0	35.0	45.0	50.0	18,100	905.0
Maize-Sorghum	-	15.0	20.0	20.0	25.0	10,900	271.5
Specialist Vegetables	-	7.5	9.0	11.0	12.0	5,240	91.2
Non-specialist Vegetables	-	7.5	9.0	11.0	12.0	3,380	47.0

8. Maxipak wheat varieties are expected to grow in the area with irrigation and proper fertilizer application. Under experimental conditions high yields of more than 4 tons/ha are obtained; however, as a project area average 3 tons/ha are projected at the 15th year. Barley would produce higher yields after leaching, but would be replaced by more profitable wheat. Linseed and sesame are the main oil-producing crops in the project. Although fusarium disease in sesame is at present a limiting factor, it would be controlled with resistant varieties. Groundnuts would be among new crops in the project area. Several varieties have been tried on an experimental basis and these trials will continue. A gradual introduction of groundnuts is anticipated, and in the early project years, substitution with other summer crops, except cotton, is possible. Cotton would be an important crop in the summer, but water supply would be a major limiting factor. Cotton in the project area at present is an insignificant crop with a very low yield of about 800 kg seed-cotton per hectare, mainly due to the lack of irrigation. With irrigation, mechanized cultivation, proper seeds, fertilizer and pesticides, as well as necessary extension support under the project, a yield of 3 tons/ha would be achieved at full development. Pulses are well known in the project and achieving the ultimate yield of 1.2 ton/ha would present no difficulties. Maize, on the other hand, would require more attention since hybrid maize is planned in the project. Yields of 3 tons/ha at full development are projected.

9. Some 1,440 ha are planned for citrus, predominantly Valencia oranges. This industry would require a thorough planning from the start as it will take time to produce some 600,000 required planting materials in nurseries. Although climatologically the Lower Khalis area is suitable for citrus growing, trees require some protection from frost damage in winter and excessive desiccation during summer. Citrus is currently planted under date palms. The project, however, plans for shelterbelts of casuarina trees. Citrus will be interplanted with vegetables during the early years.

10. Fodder crops would be devoted primarily to sheep production, which is familiar to local farmers. Fodder cropping can be organized with alfalfa as a perennial crop on one unit of land, and berseem-barley mixture in winter and maize-sorghum mixture in summer on another unit of land, allowing occasional rotation between alfalfa land and berseem-barley land. However, it is assumed that fodder for sheep would be produced on the basis of a 100% berseem-barley mixture in winter and a 60% maize-sorghum mixture in summer, with projected per hectare yield increases from 30 to 50 tons, and 15 to 25 tons, respectively. Attainment of these yields is considered realistic on the basis of current practice. Since the present primitive vegetable farming provides no basis for yield projection, the potential yield, based on soil and climatic conditions as well as the current vegetable yields in Cyprus, was used as a conservative full development yield level which is estimated at the weighted average of 12 tons/ha.

Farm Input

11. Labor requirements and availability with and without the project are presented in Figure 1. The project as a whole would not require any outside labor and most farm models would be able to handle their farming within their normal family labor, except for minor labor shortages for vegetable farms in one summer month which would be offset by child labor and additional working hours. These shortages are not costed in the farm budgets as there would be no additional cash costs. Citrus farms are expected to incur an average of 25 man-day monthly labor shortages in February, March and April, at full development. This shortage is costed in the citrus farm budget at market wage rate, assuming the annual shortage would begin to occur in 1984, increasing to the full development shortage proportionally to the citrus output.

12. Together with irrigation, drainage and extension services which would be provided as direct project activities, the attainment of the projected agricultural production would require significantly increased application of agricultural inputs. Important items include seed, fertilizer and agro-chemicals, machinery and credit. Most of the seeds except certain vegetable seeds, groundnuts and hybrid maize, will be domestically supplied. Fertilizer requirements, which may be close to 100,000 tons at full maturity, and other chemicals, including insecticides and weedicides, may have to be imported. Table 5 summarizes fertilizer requirements by each crop in the project, at full development.

Table 5: FERTILIZER REQUIREMENTS[1]

Crop	Ammonium Sulphate (kg/ha)	Triple Super Phosphate (kg/ha)
Wheat	400	100
Barley	400	100
Linseed	200	-
Broad Beans	25	50
Winter Pulses	25	50
Groundnuts	25	50
Cotton	400	250
Sesame	200	-
Maize	450	100
Sorghum	400	100
Citrus	750-1,000	100
Solanaceous and Leafy Vegetables	400-600	100-150
Brassica	200-300	100-150
Alfalfa	-	50-100
Forage Mixtures	250-350	

[1] No potash required.

13. Since farming in the project area would be heavily mechanized, a great deal of farm machinery would be required, mainly tractors and combine harvesters. These machines are now widely used in Iraq and there are already some 140 tractors in the project area. Ownership and operation of farm machinery would be left to farmers and contractors while the credits for farm machinery would be provided by the Agricultural Bank. Although maintenance of these machines would be in private hands, the project Administration would assist in the supply of spare parts and operation of repair stations.

14. Credits for purchase of all agricultural inputs would be made available through the Agricultural Bank. While these credits are essential for production, the costs are relatively small. Table 6 shows estimated aggregate credit requirements and their costs.

15. Table 7 shows the estimated cost of production inputs per ha at full development, but in current value, excluding family labor, water charges and other taxes. Farm machinery costs were estimated on a constant basis. Other production costs for the years prior to full development were estimated by linearly extrapolating the difference between the costs at the first year and at full development on the basis of:

30% of differentials at year 5
60% of differentials at year 10
100% of differentials at year 15

Table 6: ESTIMATED PROJECT CREDIT REQUIREMENTS AND COSTS

Year	Credit Requirements 1/ (ID million)	Costs of Credits 2/ ('000 ID)
1975	0.1	3.0
1976	0.9	27.0
1977	1.6	48.0
1978	2.0	60.0
1979	2.1	63.0
1980	1.9	59.0
1981	2.0	60.0
1982	2.2	66.0
1983	2.4	72.0
1984	2.4	72.0
1985	1.5	45.0
1986	1.6	48.0
1987	1.7	51.0
1988	1.7	51.0
1989	1.8	54.0

1/ Based on 100% of input costs for 1974 through 1979, 80% for 1980-85 and 50% thereafter.
2/ Based on 3% annual interest rate.

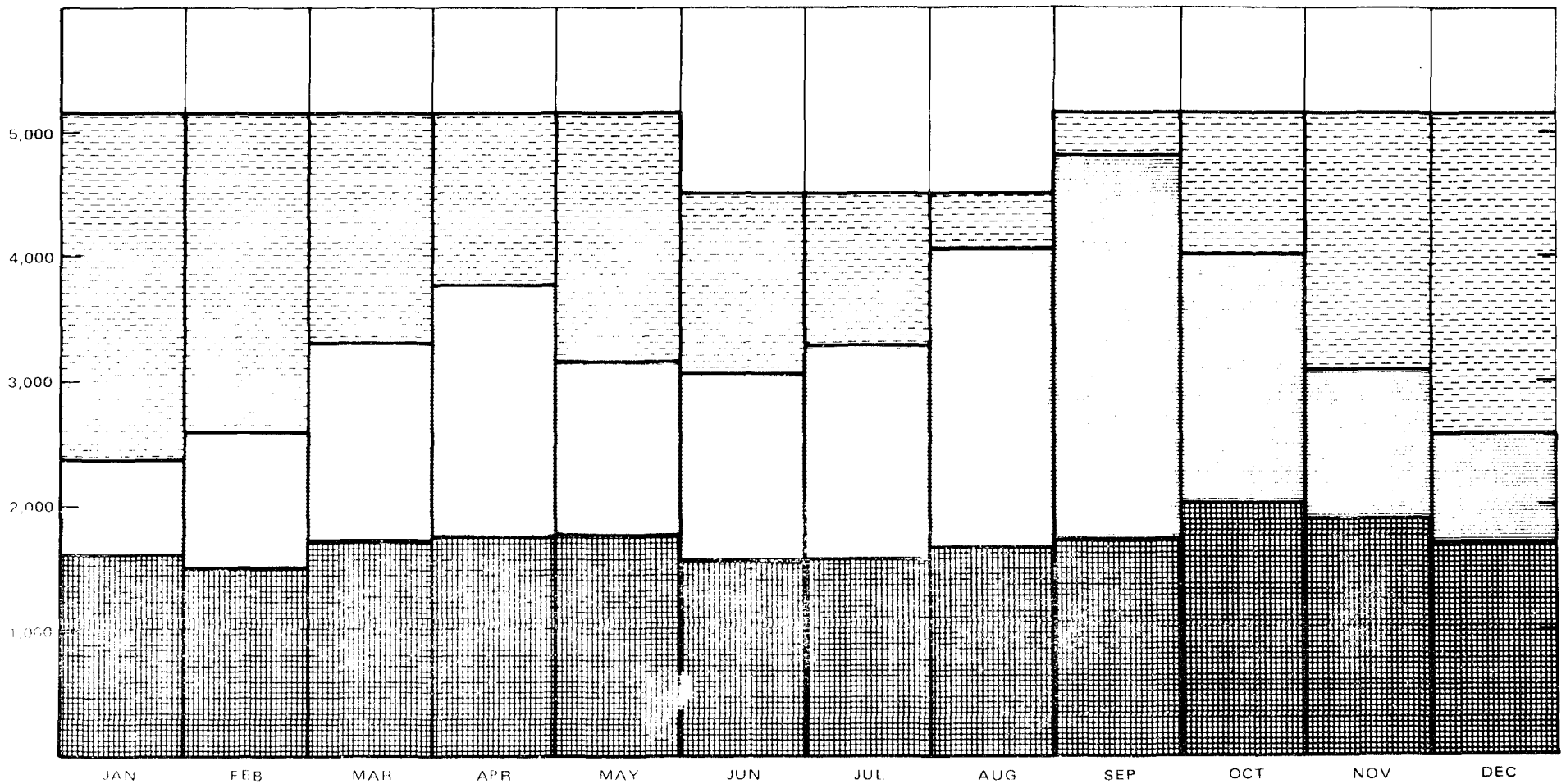
Table 7: COSTS OF INPUTS AT MATURITY

Crop	Seeds, Fertilizer Pesticides and Weedicides	Machine Hire	Other Costs 2/	Total
	(ID/ha)			
Wheat	21.0	11.2	3.2	35.4
Barley	18.8	11.2	3.0	33.0
Linseed	5.6	8.4	1.4	15.4
Broad Beans	10.1	10.8	2.1	23.0
Winter Pulses	8.5	10.0	1.9	20.4
Groundnuts	7.4	8.4	1.6	17.4
Cotton	34.7	13.6	4.8	53.1
Sesame	5.9	7.6	1.4	14.9
Maize (hybrid)	20.3	12.4	3.3	36.0
Sorghum	13.0	11.2	2.4	26.6
Vegetables	66.0	20.0	8.6	94.6
Citrus	73.0 3/	23.0	7.0	103.0
Forage	28.0	17.0	5.0	50.0

1/ Does not include costs for labor, taxes, water charges.
2/ Includes miscellaneous items and contingencies.
3/ Includes ID 26 of amortized annual costs for seedling trees.

IRAQ
LOWER KHALIS IRRIGATION PROJECT

LABOR UTILIZATION WITH AND WITHOUT PROJECT
(in thousand man-hours)



Potential supply of labor.

Incremental labor utilization attributable to Project.

Labor utilization without Project.

Bases of Estimates

(1) 25 working days/month, 8 hours/day, except in June, July and August, in which only 7 hours/day.
(2) Normal working population comprises ages 15 59.
(3) Project area population is based on 1975 figures which are estimated by applying 3.3% annual growth rate to the 1970 population.

World Bank - 6692

ANNEX 10
FIGURE 1

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Market Prospects and Prices

1. The projected incremental agricultural output from the project would encounter no market constraints, since it would be small relative to estimated total future domestic requirements. Because the project is located near Baghdad City the market prospects for high value crops such as vegetables and citrus are favorable. The quantity of the project's agricultural output to enter international markets is expected to be very small relative to the total world trade volume, and therefore, its effect on world market prices would be negligible.

2. The domestic consumption items include: legume seed crops (broad beans and winter pulses), vegetables, citrus, and livestock output. A small quantity of barley would be produced only during the early project years. The project's economic benefits from domestically consumed items are estimated on the basis of actual price projections exclusive of inflationary trends, adjusted to the farmgate price. The exception is citrus, for which the expected world market prices, adjusted to farmgate values, have been used because of the unusual domestic citrus prices in Iraq due to the import ban.

3. Traded items include cereals (wheat, maize and barley), cotton and oilseeds (linseed, sesame and groundnuts). Whether or not these are actually exported, economic benefits were estimated by shadow-pricing them at 1980 long-term world market prices, excluding inflationary effects, adjusted to the farmgate values to reflect the quality differentials and freight, transportation, handling and other costs, as may be applicable between respective international markets to Basrah port, Basrah port to Baghdad, and Baghdad to farmgate. The 1980 world market prices were provided by IBRD's Economics Department.

4. Financial prices of commodities have been derived by deducting transportation, handling and other costs between the project area and wholesale markets in Baghdad. Where the first crop year's price projection is higher than the 1980 price, the difference is gradually reduced, assuming that by 1980 the level of domestic prices of all commodites, except citrus, would be in parity with that of the world market.

5. The domestic price projections of agricultural commodities have been based primarily on the general price trend in the Baghdad wholesale market in the past 4-5 years and future supply and demand considerations. These estimates, however, are crude because they required considerable judgment and common sense in the absence of reliable and systematic quantitative information on historical prices, past and future consumption/production trends, and price elasticity of demand as well as income elasticity of demand

for various agricultural commodities in Iraq. The detailed public investment programs in the agricultural sector following the recent national emphasis on agricultural expansion, which would undoubtedly have an important bearing on the general agricultural output, markets and prices, are not as yet firmed.

6. The farmgate prices of project output used in both financial and economic analysis are presented in Table 1. Prices of agricultural inputs used for financial and economic analysis, including fertilizer, seeds, pesticides and hired labor, are based on the current market prices in the project area. The basic data on production inputs are presented in Annex 10.

Cereals

7. Iraq had been exporting wheat until the early sixties, but in recent years has become an importer. However, the Government hopes Iraq will become a net exporter of wheat by the mid 1970's. The farmgate price of ID 24/ton has been used for both economic and financial analysis for the entire project period. This price was derived by deducting ID 2/ton from ID 26, the projected 1980 Persian Gulf cif Mexipak wheat price per ton (US$72/ton before currency realignment), to account for transportation, handling and other costs between port and farmgate via Baghdad.

8. The maize production from the project is expected to reach about 4,000 tons at full maturity. Maize would be used mainly as poultry feed without entering the international market, and its financial prices have been based on domestic price projections, adjusted to the farmgate value. However, the economic benefits from maize have been evaluated at the farmgate price of ID 20/ton derived by deducting ID 3/ton for inferior quality differentials, transportation, handling and other costs from the projected 1980 Persian Gulf cif per ton maize price of good quality American corn.

9. Both financial and economic prices of barley, which would appear only during the early project years in small quantity, are based on the actual farmgate price projection of ID 14/ton. This price is comparable to the long-term international market of a similar quality of barley, when adjusted to the farmgate price.

Oilseeds

10. Government places emphasis on expansion of oilseed production and processing in an effort to achieve oilseed self-sufficiency in the face of anticipated rapid increase in demand. The farmgate per ton prices for linseed and groundnuts (unshelled) are assumed to fall gradually from ID 38 and 42, respectively, in 1974 to the farmgate adjusted international price of ID 36. The unshelled groundnut price of ID 36/ton is arrived at by deducting all costs to the farmgate and decortication costs of ID 2.9/ton from the projected 1980 cif Europe price of ID 64/ton of shelled groundnuts. The ratio of unshelled nuts to shelled nuts by weight was based at 1 to 0.7. The farmgate adjusted international price of linseed is estimated by allowing ID 7 for freight and all other costs to the farmgate from the projected 1980 Europe cif linseed price of ID 43/ton.

Cotton

11. The total project output of seed cotton at full maturity would be about 17,000 tons of the medium staple variety (Coker 100 wilt), which would be a small proportion of world cotton trade volume. The farmgate cotton price of ID 76, which has been used in both financial and economic evaluations, has been derived on the following bases:

(a) Projected 1980 cif Liverpool price of Coker 100 wilt: ID 197/ton of lint.

(b) Lint value per ton seed cotton: ID 69, assuming one ton of seed cotton consists of 350 kg of lint and 650 kg of seed.

(c) Farmgate value of cottonseed content per ton of seed cotton after deducting handling, transportation and other costs: ID 15.

(d) Freight, transportation, handling, ginning and other costs per ton of cotton between farmgate and Liverpool: ID 8.

The farmgate price = (b) + (c) - (d) = ID 76.

Vegetables and Citrus

12. Citrus and vegetables would be among the most important crops in the project, and their output at full maturity would reach some 36,000 tons and 138,000 tons, respectively. These quantities would contribute about 15-20% of the national citrus requirements and 5% of the national vegetable consumption (or 20% of the Baghdad area's). Marketing and transportation would pose no problem because of easy access to Baghdad City. Marketing of fruits, including citrus, is currently in the hands of private wholesalers in Baghdad who undertake direct buying and selling. Vegetables are also sold in the privately owned wholesale vegetable markets, but directly by the growers. The current fruit and vegetable market structures and operations are quite primitive and inefficient. In view of the anticipated increase in consumption and supply of fruits and vegetables, Government is seriously considering a direct participation in these markets under modernized organization, including storage, packing and quality grading facilities which would substantially reduce the current losses and price fluctuations.

13. The farmgate vegetable price of ID 25/ton was used for both economic and financial analysis, assuming that the weighted average annual wholesale prices of vegetables between 1966 and 1968, which have been 20% higher than the above farmgate prices, would continue in the future, except for cauliflower which is expected to decline by 25-35% by 1986. This general price assumption was made in view of:

(a) high income elasticity of demand for vegetables and Government's policy to encourage vegetable consumption;

(b) the relatively small proportion of the project vegetable output in relation to the total consumption requirements in the Baghdad area or in the country; and

(c) Government's intention to improve vegetable marketing in the Baghdad area.

14. Citrus prices are extremely high in Iraq, compared to other countries, reflecting the shortage in supply primarily due to the import ban on citrus which was reportedly undertaken in order to prevent the Mediterranean fruit fly, "ceratitis capitate." The weighted seasonal average prices of oranges in Iraq between 1967-1971 has been in the neighborhood of ID 100/ton.

15. The future prices of citrus in Iraq cannot be projected with certainty because they would depend mainly on (a) the degree and pace of Government's citrus self-sufficiency plan; (b) per capita consumption of citrus and income elasticity of demand for citrus; and (c) price elasticity of demand for citrus. The current annual citrus production is estimated at 8,000-10,000 tons and the self-sufficiency annual requirement in twenty years would range somewhere between 150,000-300,000 tons, depending on projected per capita citrus consumption. Although Government hopes to achieve domestic citrus self-sufficiency, there is no detailed plan other than various studies currently being undertaken, including UNDP/FAO Fruits and Vegetables Production and Marketing Project. The citrus component of this project is a part of Government's self-sufficiency drive. In view of the proportion of this project's output relative to the national citrus requirements and specialized public efforts needed in developing the citrus industry, project citrus farmers are likely to enjoy extraordinary benefits.

16. The level of citrus price used in economic analysis has some effect on economic rate of return though it is not a critical determinant of the project's economic viability. In the initial economic rate of return calculation ID 20/ton was used as the most conservative farmgate-adjusted international price of Valencia, which is the orange brand to be grown in the project, even if there were no import restriction on citrus in Iraq. This is about 15-25% lower than the average wholesale market price of similar grade oranges in major orange production centers in Spain. In the sensitivity analysis alternative prices of linearally and exponentially declining trends from the current level to the ID 20/ton in 1988 have been used to test their effects on the rate of return.

17. Since the financial prices of citrus would have a significant bearing on farmers' income and repayment capacity, an attempt was made to project the actual domestic farmgate price of citrus, assuming that the current citrus import ban and the stated Government intention to achieve citrus self-sufficiency are permanent measures. The linearally declining citrus prices from the current level of ID 100/ton to ID 20/ton in 1988 have been used for financial evaluation. This would be roughly consistent with the projected price trend under the assumptions:

(a) 3% per annum growth rate of population and income;

(b) 0.3 income elasticity of demand for citrus and 1.4 price elasticity of demand for citrus;

(c) citrus self-sufficiency output to be achieved at 150,000 tons in 1988 from the current annual output of 8,200 tons.

Legume Seed Crops

18. Production of winter pulses and broad beans is expected to reach about 3,400 and 5,700 tons, respectively, at full development. In view of the projected large increase in national requirements, the marketing of these crops would pose no problem. Since the current market for these crops is very small in Iraq, available market information is insufficient for use as the basis of future price projection. The farmgate price of ID 32/ton was used for both economic and financial analyses of broad beans. Declining per ton financial prices for winter pulses, ID 58/ton in 1974 and ID 53/ton in 1979, were used while the economic analysis used a constant price of ID 53/ton.

Livestock Products

19. Over 18,000 ha of the project's cropped area would be devoted to fodder production. Since there is no meaningful market for forage in Iraq, the economic benefits and farm incomes were indirectly attributed to sheep production. Although a small proportion of forage is likely to be used for dairy, which may bring a slightly higher income, this difference was ignored for computational convenience. There has been shortage of mutton and milk in Iraq throughout the past decade and this is reflected in the price trends which doubled between 1960-1968. This trend is expected to continue for some time to come as the rate of increase in demand is likely to outpace that of production. The following farmgate prices have been used for both economic and financial analyses:

sheep and lambs	ID 7.2 per head
milk	50 fils (ID 0.05) per kg
wool	200 fils (ID 0.2) per kg

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Market Prospects and Prices

FARMGATE PRICES USED FOR ECONOMIC AND FINANCIAL ANALYSIS OVER PROJECT PERIOD [1]

Year	1974	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	2022
Commodity	(ID per metric ton unless otherwise indicated)															
Wheat	24/30	24/29	24/28	24/27	24/26	24/25	24/24	24/24	24/24	24/24	24/24	24/24	24/24	24/24	24/24	24/24
Linseed	36/38	36/38	36/37	36/37	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36
Broad Beans	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32	32
Winter Pulses	53/56	53/55	53/55	53/54	53/54	53/53	53/53	53/53	53/53	53/53	53/53	53/53	53/53	53/53	53/53	53/53
Sesame	71	71	71	71	71	71	71	71	71	71	71	71	71	71	71	71
Maize	20/26	20/25	20/24	20/23	20/22	20/21	20/20	20/20	20/20	20/20	20/20	20/20	20/20	20/20	20/20	20/20
Cotton	76	76	76	76	76	76	76	76	76	76	76	76	76	76	76	76
Groundnuts	36/42	36/41	36/40	36/39	36/38	36/37	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36	36/36
Citrus	20/82	20/78	20/73	20/69	20/64	20/60	20/55	20/51	20/46	20/42	20/37	20/33	20/28	20/24	20/20	20/20
Vegetables [2]	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
Barley	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14	14
Milk (kg)	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Wool (kg)	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Sheep or lamb (head)	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2	7.2

1/ Where prices used for economic and financial analysis differ, economic is above line and financial below.

2/ Weighted average prices of tomatoes, potatoes, eggplant, green peppers, green beans, green cowpeas, okra, cucumber.

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ANNEX II
Table 1

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Farm Budget and Repayment Capacity

1. Representative farm budgets for typical project farm types are presented in the attached Tables 1 through 4. Bases of input and output estimates and prices are presented in Annexes 10 and 11.

2. These are partial farm budgets in the strict sense as they do not include (a) minor income which farmers may earn from off-farm activities or on-farm wage earnings; and (b) small livestock income of about ID 15 per family attributable to the crudely estimated 5,000 cows and 13,500 goats supported by the wasteland and fallow land in the project command area which were assumed to be unaffected by the project.

3. Family labor input has not been costed in the farm budgets; therefore, net revenues in farm budgets are inclusive of gross returns from family labor. Minor labor shortage which does not exceed 10% of the normal labor availability of the respective farm type has not been costed, unless the shortage continues longer than one month. It was assumed this minor shortage would be offset by child labor or slightly longer working hours. Citrus farms are expected to face some labor shortages in the months of February through April from 1984 on. These shortages are costed at ID 0.5 per man-day.

4. Illustrative repayment schedules aimed at recovering the entire investments and O & M costs over the lifetime of the project, at 3% interest rate, have been included in the attached farm budget tables. The repayment schedules are progressive and somewhat redistributive, and therefore, such schedules would need to be applied in the context of general agricultural taxation. Since the Government's planning and policies would become a major determinant of levels and distribution of incomes of the farmers in the project area, some redistributive capital recovery would be justifiable. For example, the projected citrus income stream includes a present value of some ID 1.0 million incremental income solely attributable to the protected price from the Government's citrus import ban. The following table illustrates the magnitude of difference in various methods of assessing project charges at full development. It also reveals that the project charges proportional to land areas would be most regressive.

Full Development Project Charges Levied with Alternative Criteria[1]

Alternative Criteria	Farm Type — 4 ha Mixed Arable	8 ha Mixed Arable	Vegetable Specialist	Citrus Specialist
	ID			
Progressive charges in this analysis	68 (17%)	228 (32%)	279 (35%)	370 (39%)
Proportional to area	152 (37%	304 (42%)	137 (17%)	113 (12%)
Proportional to water consumption	147 (35%)	265 (37%)	189 (24%)	189 (20%)

[1] Figures in brackets represent percentage of farm income levied as project charge.

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LOWER KHALIS IRRIGATION PROJECT

FARM BUDGET AND REPAYMENT CAPACITY: SPECIALIST CITRUS FARM (3 HA)

	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991
	(Iraqi Dinars)															
Gross Revenue at Farmgate																
2 ha Citrus	-	-	-	128	360	770	1,020	1,104	1,260	1,332	1,320	1,176	1,056	920	960	1,000
2 ha Vegetables intercropped	308	284	226	240	256	274	190	100	-	-	-	-	-	-	-	-
0.5 ha Vegetables	88	106	113	125	138	150	162	169	175	188	194	200	206	212	219	219
0.5 ha Forage/Sheep	47	49	51	53	55	58	60	62	64	67	69	71	74	76	78	78
Total	443	439	390	546	809	1,252	1,432	1,435	1,499	1,587	1,583	1,447	1,336	1,208	1,257	1,297
Production Cost (excl. family labor)																
2 ha Citrus	18	34	52	70	82	98	130	146	164	176	186	198	206	206	206	206
2 ha Vegetables Intercropped	104	90	72	76	80	84	58	30	-	-	-	-	-	-	-	-
0.5 ha Vegetables	28	30	32	34	35	37	38	40	40	42	43	44	45	46	48	48
0.5 ha Forage/Sheep	27	28	29	30	31	32	34	35	36	39	40	41	43	44	46	46
Subtotal	177	182	185	210	228	251	260	251	240	257	269	283	294	296	300	300
Hired Labor Costs [1]	-	-	-	-	-	-	-	-	-	-	6	12	18	24	30	37
Total	177	182	185	210	228	251	260	251	240	257	275	295	312	320	330	337
Net Farm Income	266	257	205	336	581	1,001	1,172	1,184	1,259	1,330	1,308	1,152	1,024	888	927	960
National Average Farm Income [2]	197	203	209	215	221	228	235	242	249	256	264	272	280	288	297	306
Taxable Surplus [3]	69	54	-	121	360	773	937	942	1,010	1,074	1,044	880	744	600	630	656
Project Charges [4]	39	30	-	68	203	436	529	532	571	606	589	497	421	339	356	370
Disposable Family Income after Project Charges	227	227	205	268	378	565	643	652	688	724	719	655	603	549	571	590

1/ 0.5 ID per man-day. Based on average February to April full development labor shortage to be at 200 man-hours per month and earlier years' labor requirement proportional to citrus output.

2/ 1970 value estimated on the basis of per capita agricultural income and a family size of 5. Thereafter projected at a growth rate of 3% per annum.

3/ Income in excess of national average farm income.

4/ This would recover the entire investment and O&M costs over the lifetime of the project at 3% interest rate.

ANNEX 12
Table 1

April 17, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

FARM BUDGET AND REPAYMENT CAPACITY: SPECIALIST VEGETABLE FARM (3.6 HA)

	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
	(Iraqi Dinars)														
Gross Revenue at Farmgate															
2 ha Vegetables	350	424	450	500	550	600	650	676	700	750	776	800	824	850	876
1.6 ha Forage/Sheep	149	157	163	170	176	184	190	198	206	216	222	229	237	243	248
Total	499	581	613	670	726	784	840	874	906	966	998	1,029	1,061	1,093	1,124
Production Cost (excl. family labor)															
2 ha Vegetables	114	122	128	134	140	148	154	158	162	168	172	176	180	184	190
1.6 ha Forage/Sheep	87	90	92	97	100	104	108	112	118	125	127	132	136	139	144
Total	201	212	220	231	240	252	262	270	280	293	299	308	316	323	334
Net Farm Income	298	369	393	439	486	532	578	604	626	673	699	721	745	770	790
National Average Farm Income 1/	197	203	209	215	221	228	235	242	249	256	264	272	280	288	297
Taxable Surplus 2/	101	166	184	224	265	304	343	362	377	417	435	449	465	482	493
Project Charges 3/	57	93	105	127	150	172	194	204	213	235	245	253	262	272	279
Disposable Family Income after Project Charges	241	276	288	312	336	360	384	400	413	438	454	468	483	498	511

1/ 1970 value estimated from per capita income in agriculture on a family size of 5; thereafter projected at a growth rate of 3% per annum.
2/ Income in excess of national average farm income.
3/ This would recover the entire investment and O&M costs over the lifetime of the project at 3% interest rate.

ANNEX 12
Table 2

April 17, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

FARM BUDGET AND REPAYMENT CAPACITY: MIXED ARABLE AND LIVESTOCK FARM (8 HA)

	1976	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991
	(Iraqi Dinars)														
Gross Revenue at Farmgate															
5 ha Arable Field Crops	116	133	150	231	270	336	367	401	433	482	510	530	552	573	594
0.5 ha Vegetables	70	85	101	112	123	120	130	135	140	150	155	160	165	170	175
2.5 ha Forage/Sheep	233	245	255	265	275	288	298	310	323	338	348	358	370	380	388
Straw Benefits	4	5	5	8	7	9	10	10	11	13	13	14	14	15	15
Total	423	468	511	616	675	753	805	856	906	983	1,025	1,062	1,101	1,138	1,173
Production Cost (excl. family labor)															
5 ha Arable Field Crops	46	55	66	95	110	135	146	154	161	173	182	185	190	194	199
0.5 ha Vegetables	23	24	26	27	28	30	31	32	32	34	34	35	36	37	38
2.5 ha Forage/Sheep	135	140	145	150	154	163	170	175	185	195	200	205	213	218	225
Total	204	219	237	272	292	328	347	361	378	402	416	425	439	449	462
Net Farm Income	219	249	274	343	383	426	458	496	528	581	609	637	663	689	711
National Average Farm Income 1/	203	209	215	221	228	235	242	249	256	264	272	280	288	297	306
Taxable Surplus 2/	16	40	59	122	155	191	216	247	272	317	337	357	375	392	405
Project Charges 3/	8	22	34	69	96	108	122	140	154	179	191	201	211	221	228
Disposable Family Income after Project Charges	211	227	240	274	287	318	336	356	374	402	418	436	452	468	483

1/ 1970 value estimated from per capita income in agriculture on the basis of a family size of five; thereafter projected at growth rate of 3% per annum.

2/ Income in excess of national average farm income.

3/ This would recover the entire investment costs over the lifetime of the project at 3% interest rate.

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ANNEX 12
Table 3

IRAQ

LOWER KHALIS IRRIGATION PROJECT

FARM BUDGET AND REPAYMENT CAPACITY: MIXED ARABLE AND LIVESTOCK FARM (4 HA)

	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
	(Iraqi Dinars)														
Gross Revenue at Farmgate															
2.5 ha Arable Field Crops	60	69	79	120	140	171	183	201	217	241	255	265	276	287	297
0.5 ha Vegetables	70	85	101	100	110	120	130	135	140	150	155	160	165	170	175
1 ha Forage/Sheep	93	98	102	106	110	115	119	124	129	135	139	143	148	152	155
Straw Benefits	2	2	3	4	4	4	5	5	6	6	6	7	7	7	8
Total	225	254	285	330	364	410	437	465	492	532	555	575	596	616	635
Production Cost (excl. family labor)															
2.5 ha Arable Field Crops	23	28	33	47	55	67	73	77	81	86	91	93	95	97	100
0.5 ha Vegetables	23	24	25	27	28	30	31	32	32	34	34	35	36	37	38
1 ha Forage/Sheep	54	56	58	60	63	65	68	70	74	78	80	82	85	87	90
Total	100	108	116	134	146	162	172	179	187	198	205	210	216	221	228
Net Farm Income	125	146	169	196	218	248	265	286	305	334	350	365	380	395	407
National Average Farm Income 1/	191	197	203	209	215	221	228	235	242	249	256	264	272	280	288
Taxable Surplus 2/	-	-	-	-	3	24	38	51	63	85	94	101	108	115	119
Project Charges 3/	-	-	-	-	2	15	22	29	35	47	54	57	61	64	68
Disposable Family Income after Project Charges	125	146	169	196	216	233	243	257	270	287	296	308	319	331	339

1/ 1970 value estimated from per capita income in agriculture on the basis of a family size of five; thereafter projected at growth rate of 3% per annum.
2/ Income in excess of national average farm income.
3/ This would recover the entire investment and O&M costs over the lifetime of the project at 3% interest rate.

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Annex 12
Table 4

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Economic Rate of Return

The economic rate of return from the project is estimated at 11.6%. The investment costs and incremental benefit streams are presented in the attached Appendix 1 and the bases of project inputs and outputs as well as their prices are presented in Annexes 10 and 11. Assumptions and bases of this analysis include:

(a) Economic life of the project was assumed to be 50 years.

(b) Expenditures on the Derbendi Khan dam and other joint sunk costs incurred to date were treated as sunk costs and were not included.

(c) Investment costs stream includes:

  (i) capital costs including physical contingencies and O&M costs;

  (ii) the cost of enlarging the upper reach of the Khalis main canal in 1980/81 (ID 200,000) to match project peak water requirements; and

  (iii) an additional contingency item for the proportional cost attributable to the project for a pump station on the Tigris to supplement Diyala water supplies from 1985/86 onwards (16 m^3/sec capacity; 20% of estimated cost, ID 2.2 million plus O&M).

(d) Social opportunity cost of family labor was assumed to be zero. It was also assumed that minor seasonal labor shortages expected on some individual farms would be offset by the surplus labor which has no alternative employment in the area.

(e) The analysis does not take into account the nonproject-related economic activities in the area. Also excluded from "with" and "without project" benefit/cost analysis is the proportion of the livestock population assumed to be unaffected by the project, i.e. some 5,000 cows and 13,500 goats to be supported by the waste land and fallow land mentioned in Annex 12.

(f) There is no meaningful monetary market for fodder output in Iraq and therefore direct valuation of fodder would be difficult. The benefits from forage land were estimated on the basis of the imputed economic contribution of the sheep industry to be supported by the project fodder output and roughages. Using a simple annuated benefit/cost calculation, a conservative ID 65 per ha was derived as the net forage benefit at full development. Details are presented in the attached Appendix 2.

(g) The project's economic benefits from domestically consumed items are estimated on the basis of actual price projections exclusive of inflationary trends, adjusted to farmgate prices. The exception is citrus for which expected world market prices adjusted to farmgate values have been used because of the unusually high domestic citrus prices in Iraq due to the import ban. Economic benefits from internationally traded items were estimated by shadow-pricing them at 1980 long-term world market prices and adjusted to the farmgate values to reflect quality differentials, freight, transportation, handling and other costs. (For details, see Annex 11.)

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LOWER KHALIS IRRIGATION PROJECT

ECONOMIC RATE OF RETURN
INVESTMENT COSTS AND NET BENEFIT STREAM

Project Year	Investment Costs	Incremental Net Benefits
	ID '000	
1	380	0
2	1,600	0
3	4,770	-164
4	7,490	440
5	7,490	940
6	1,960	1,332
7	720	1,668
8	820	2,124
9	820	2,649
10	720	3,045
11	720	3,356
12	720	3,636
13	960	4,031
14	960	4,393
15	730	4,654
16	730	4,882
17	730	5,114
18	730	5,324
19	730	5,443
20	730	5,452
21	730	5,452
22	730	5,452
23	780	5,452
24	870	5,452
25	820	5,452
26	860	5,452
27	820	5,452
28	730	5,452
29	1,160	5,452
30	730	5,452
31	810	5,452
32-50	730	5,452

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LOWER KHALIS IRRIGATION PROJECT

Economic Rate of Return

Basis of Forage Benefit/Cost Calculation

The per-hectare net forage benefit was estimated at ID 65 based on the following:

(a) animal per hectare at full development - determined at 18, of which one would be ram and the rest ewes on the assumption:

(i) 1 ewe/1 lamb consumption rate of dry matter = 0.8 ton/year;

(ii) each hectare forage production estimated at 13 tons of dry matter would be supplemented by 3 tons of straw to be provided mainly by the arable farming sector of the project;

(iii) deducting 10% losses in forage and roughage, the number of animal stock which can be supported by one hectare of forage land would be:

$$\frac{(13 \text{ tons} + 3 \text{ tons}) \times 0.9}{0.8 \text{ tons}} = 18$$

(b) annual benefits and costs per hectare at full development -

benefits estimated at ID 155 on the bases of:

(i) 17 ewes, each ewe bringing ID 9.1/year from ID 7.2 of lamb and cull sales, ID 1.5 of milk sales and ID 0.4 of wool;

(ii) each ram brings ID 0.4 of wool per year;

costs estimated at ID 90 for 18 animals, each of them incurring total costs of ID 5.00 for:

(i)	consumption of fodder	ID 3.12
(ii)	interest cost for borrowing one animal purchase capital at 8% interest rate	ID 0.58

(iii)	veterinary costs	ID 0.30
(iv)	concentrated material	ID 1.00

(c) benefits and costs per hectare prior to full development - calculated proportionally to the increase in animal stock which would grow proportionally to the rate of project fodder yield increase.

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LOWER KHALIS IRRIGATION PROJECT

Results of Sensitivity Tests

	Assumptions	Economic Rate of Return
(a)	The basic run excluding sunk costs	11.6
(b)	Include in the first year investment stream the joint sunk costs of ID 4.8 million, consisting of	
	(i) ID 4.0 million: 1/5 of Derbendi Khan dam costs (1962);	
	(ii) ID 0.3 million: 1/6 of Diyala weir reconstruction costs (1969); and	
	(iii) ID 0.5 million: 1/2 of Khalis canal reconstruction costs (1971)	9.6
(c)	15% overall reduction in prices or yields of arable crops	10.5
(d)	15% increase in investment costs (including on-farm development costs)	10.5
(e)	15% increase in production costs	10.2
(f)	2 years' delay in benefits	9.7
(g)	15% reduction in gross benefits	7.9
(h)	15% increase in gross benefits	15.0
(i)	Linearly declining citrus prices from ID 100/ton in 1971 to ID 20/ton in 1988	12.1
(j)	Exponentially declining citrus prices from ID 100/ton in 1971 to ID 20/ton in 1988	11.9
(k)	One-year delay in citrus planting	11.5
(l)	Vegetable prices increased to ID 28/ton	12.6
(m)	Cotton yield decreased to 2.5 ton/ha	11.1

Continued....

Assumptions (Cont'd.)	Economic Rate of Return
(n) Optimum cropping pattern obtained from linear programming which suggests replacement of low profit commodities by cotton (see Technical Note 1 below)	12.7
(o) 8% reduction in benefits from tradable commodities assuming the most pessimistic effects of currency realignment on benefits from tradable items (see Technical Note 2 below)	11.1
(p) Development Area III as an independent project	10.3
(q) Without shadow-pricing family labor by costing all labor at ID 0.5/manday.	9.0

Technical Note 1

1. A linear programming analysis was performed in order to (a) determine the optimum full development cropping pattern on the gravity-fed arable land in Development Areas I and II which would maximize net return, and (b) estimate the effects of the maximum return thus obtained on the economic rate of return from the project.

2. The impact of the optimum cropping pattern benefits on economic rate of return would result in an increase of the rate by 1.1%, i.e. 12.7% compared to the initial rate of 11.6%. The optimum cropping pattern under the most realistic assumption on labor availability would be 11,700 ha of forage and 10,020 ha of cotton, compared to the initial nonoptimum project cropping pattern of:

11,700 ha forage
4,700 ha cotton
2,340 ha sesame
1,170 ha groundnuts
1,170 ha maize

3. The following constraints have been assumed in the analysis:

(a) Forage area would be limited to 11,700 ha which was equal to the forage land in the project cropping pattern.

(b) Citrus and vegetable land areas were fixed.

(c) There would be no room to improve winter cropping patterns because they were close to satisfying equi-marginal return conditions.

(d) Peak labor would not be unlimited at zero social costs.

(e) Competing crops in the cropping pattern would be restricted to cotton, sesame, groundnuts, maize and fodder.

(f) Peak water and total annual water supply would be limited.

Technical Note 2

1. Economic benefits from tradable project output are estimated at shadow-priced farmgate value of 1980 long-term international market prices of relevant commodities. These price projections were based on the precurrency realignment estimates of Economics Department expressed in U.S. currency. While no precise effects can be foreseen at this time, Iraq's dollar revenue from tradable agricultural commodities, assuming they would be sold in the international market, would be affected and their magnitude will depend on the changes in trade patterns. This sensitivity merely tests its effects on economic rate of return assuming the extreme case where the full revaluation of the Iraqi dinar relative to the U.S. dollar (8%) would fall on the Iraqi revenue from relevant agricultural export commodities.

June 22, 1972

IRAQ

LOWER KHALIS IRRIGATION PROJECT

Economic Consideration of Pumping

1. The future water supply in the Diyala basin would be affected by Iranian water resources developments for the extension of irrigation on the headwaters and tributaries of the Diyala river, and Iraq may need to supplement the river flow by pumping from the Tigris river. Effects of pumping from the Tigris river on the economic viability of this project have been estimated under alternative assumptions regarding the level of Iranian water resources diversion and allocation of pumping costs. The results are summarized below.

2. The 11.6% initial estimate of the economic rate of return includes estimated joint costs (1/5 of the cost of a 16-m^3/sec pump installation which is estimated to cost a total of ID 2.2 million in 1985/86) of water supply contingency, assuming 10% of the median flow of the Diyala would be diverted by Iran.

3. Sensitivity Test A. The objective of this test was to estimate the required size of pump and to measure the impact of costing that pump on economic rate of return, assuming that:

(a) Iran would divert 20% of water resources;

(b) the entire Diyala basin would be developed;

(c) total basin water shortage or the entire water requirements for the Lower Khalis would be supplied by pumping from the Tigris; and

(d) the entire costs of pumping would be charged against Lower Khalis Project.

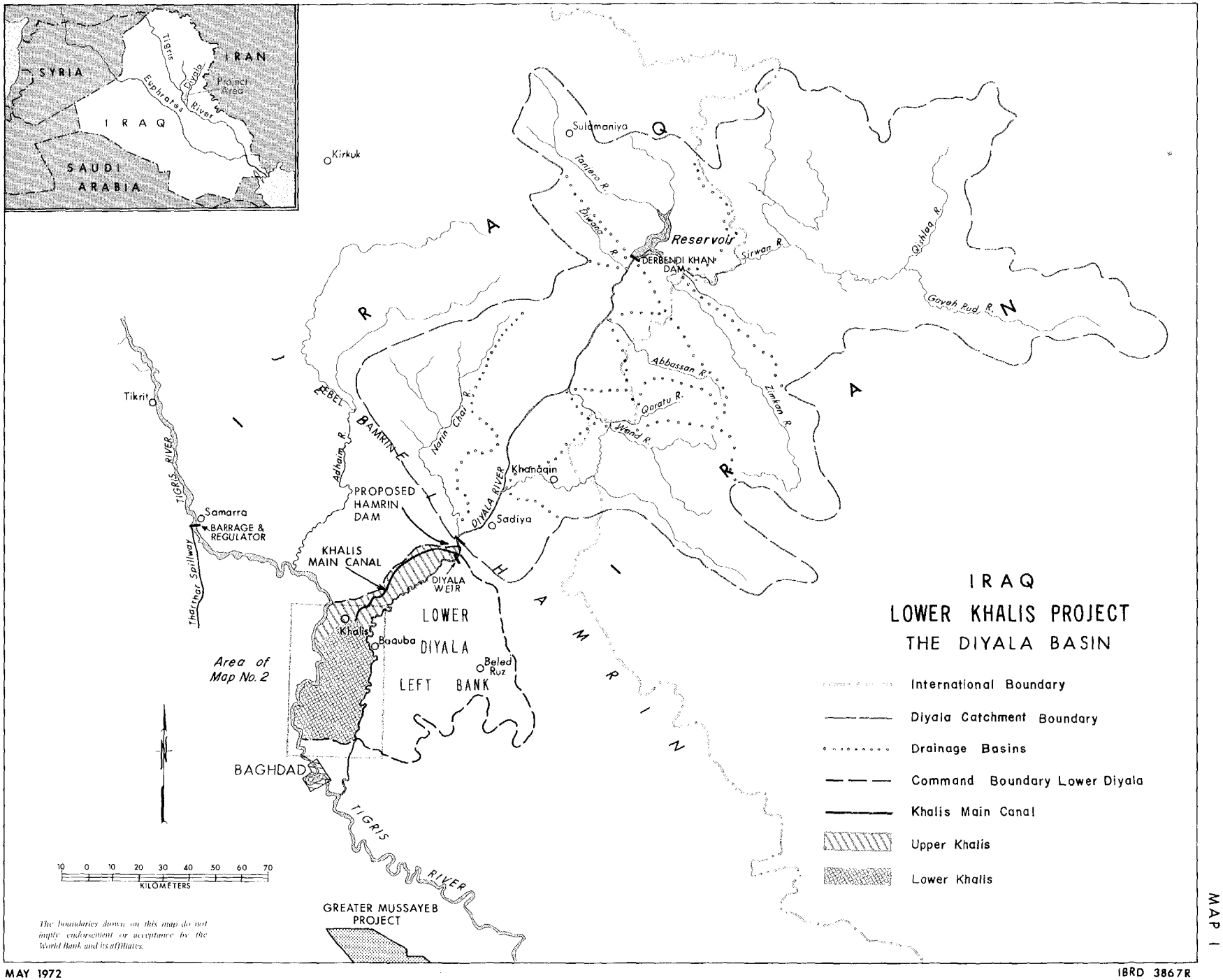
The required pump size was estimated to be 45 m^3/sec with costs of about ID 7 million in 1984 and 1985. The resulting economic rate of return was 10.7% when the total was charged against Lower Khalis. However, the rate would be 11.5% if the apportioned cost which is 20% of the total cost of ID 7 million is charged against Lower Khalis Project. If the supplementary pump installation was to be required and constructed as early as 1978/80, the rate of return would be 10%.

4. Sensitivity Test B. The objective of this test was to estimate the difference in project cost and rate of return between the proposed arrangement of minimum pump-supplied area (15% of the project), and an alternative with the pumping capability and area increased to 35% to absorb

the existing rate of pumping from the Tigris (19 m^3/sec). The net increase in cost of the alternative would be about ID 0.9 million, and the rate of return would be reduced from 12% to 11%.

5. Sensitivity Test C. The objective of this test was to (a) determine the optimum pump size which would minimize the sum of pumping costs and crop losses under the same 10% Iranian water diversion assumption, and (b) estimate the impact of the above pump and the associated crop losses on economic rate of return. The pump size turned out to be 32 m^3/sec at a cost of ID 3.9 million with associated operating costs and crop losses of about ID 1.7 million at 1985 value. The resulting economic rate of return was 10.9%.

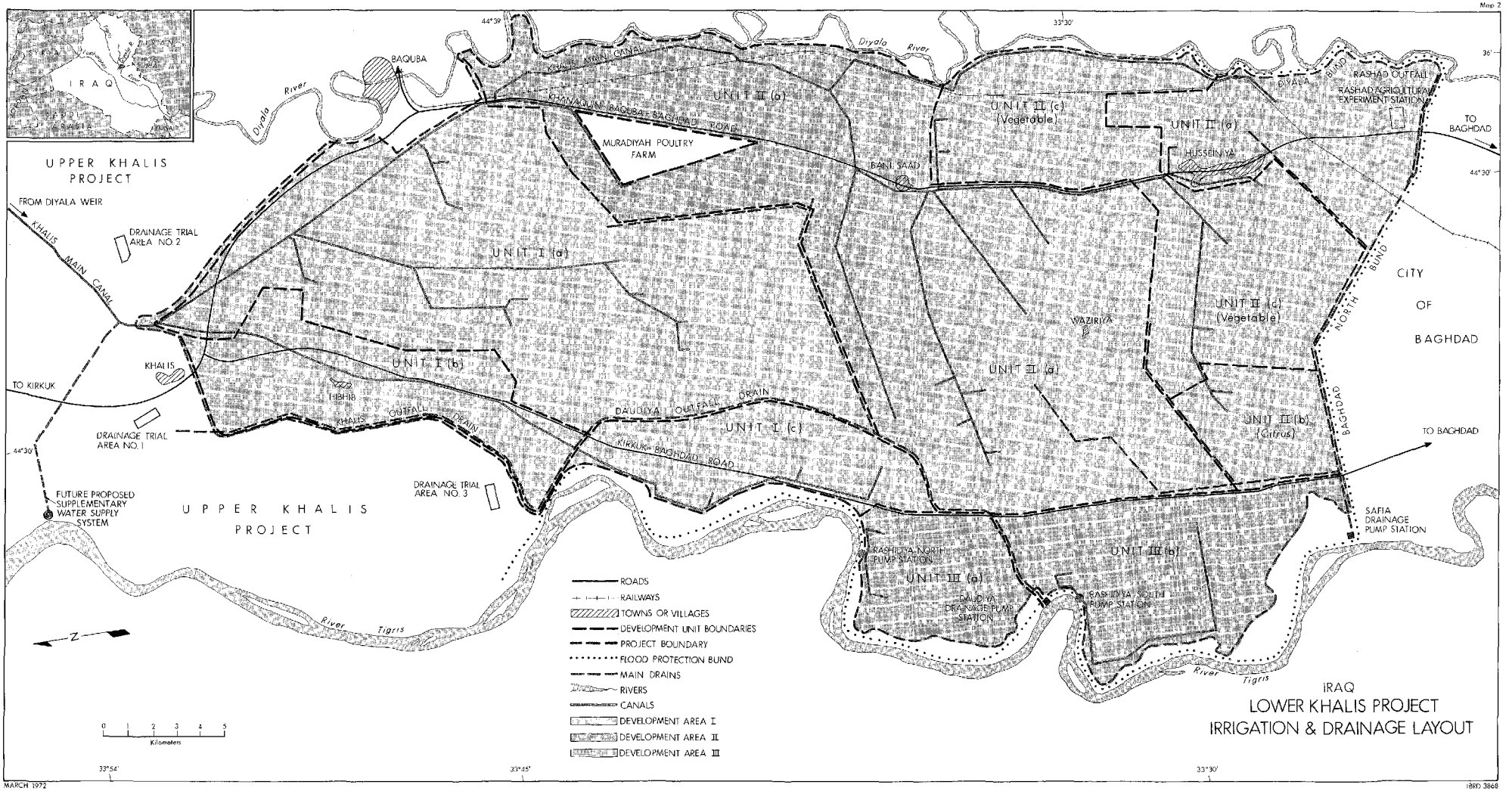
June 22, 1972



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MAP I



IRAQ
LOWER KHALIS PROJECT
IRRIGATION & DRAINAGE LAYOUT

MARCH 1972

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